

BFS Traversal

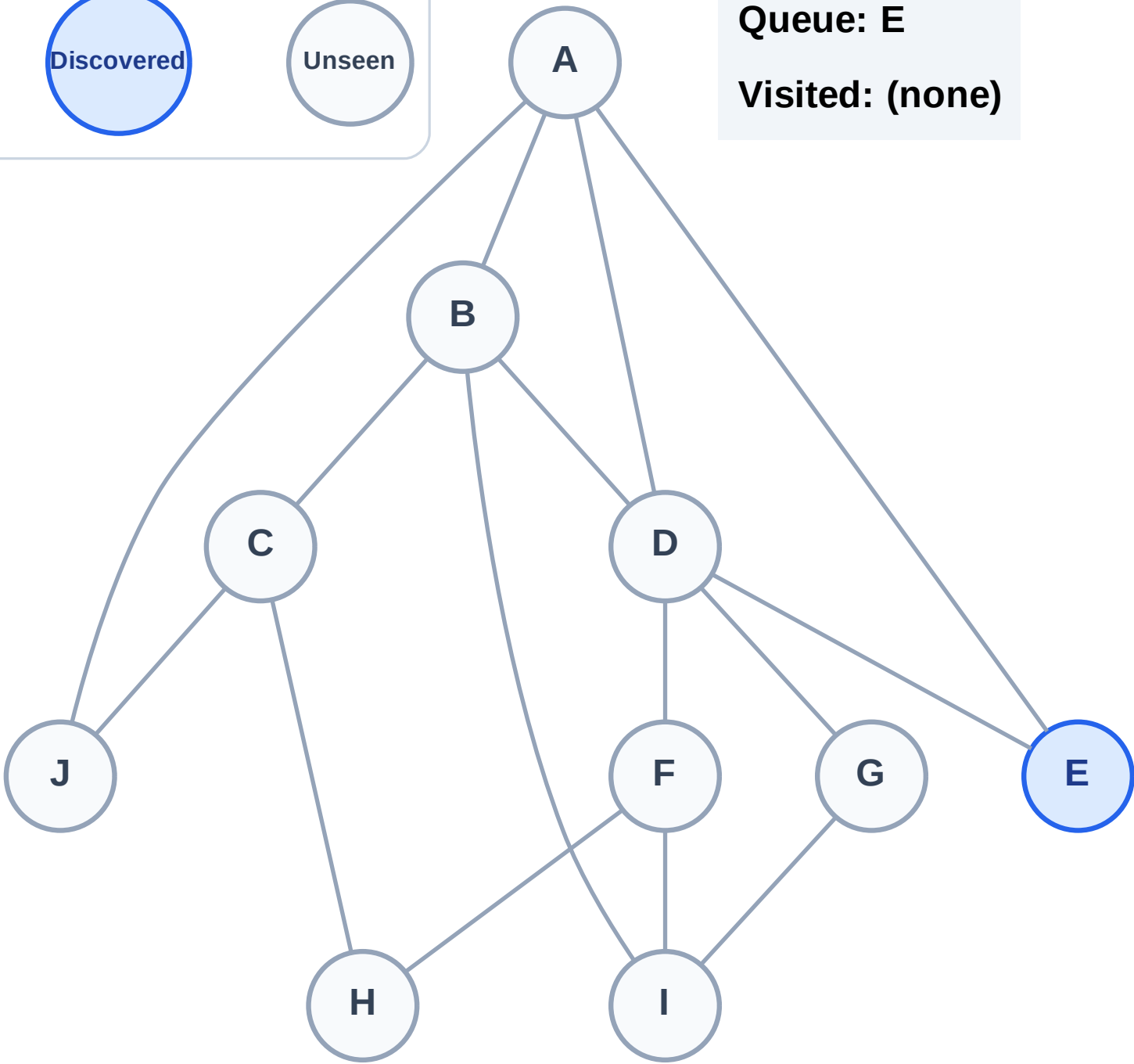
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

Complete

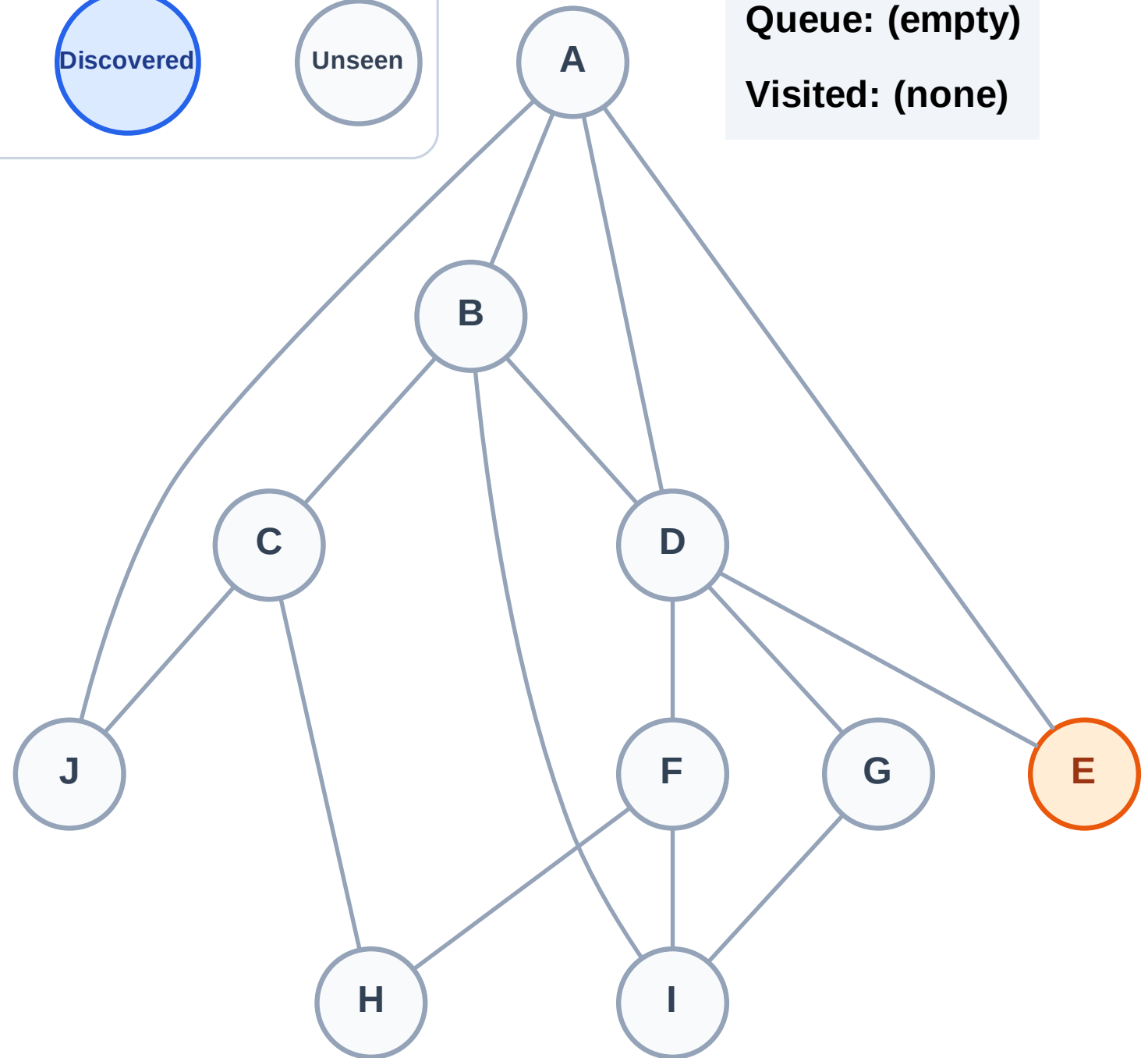
Processing

Discovered

Unseen

Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover A, D from E

Legend

Complete

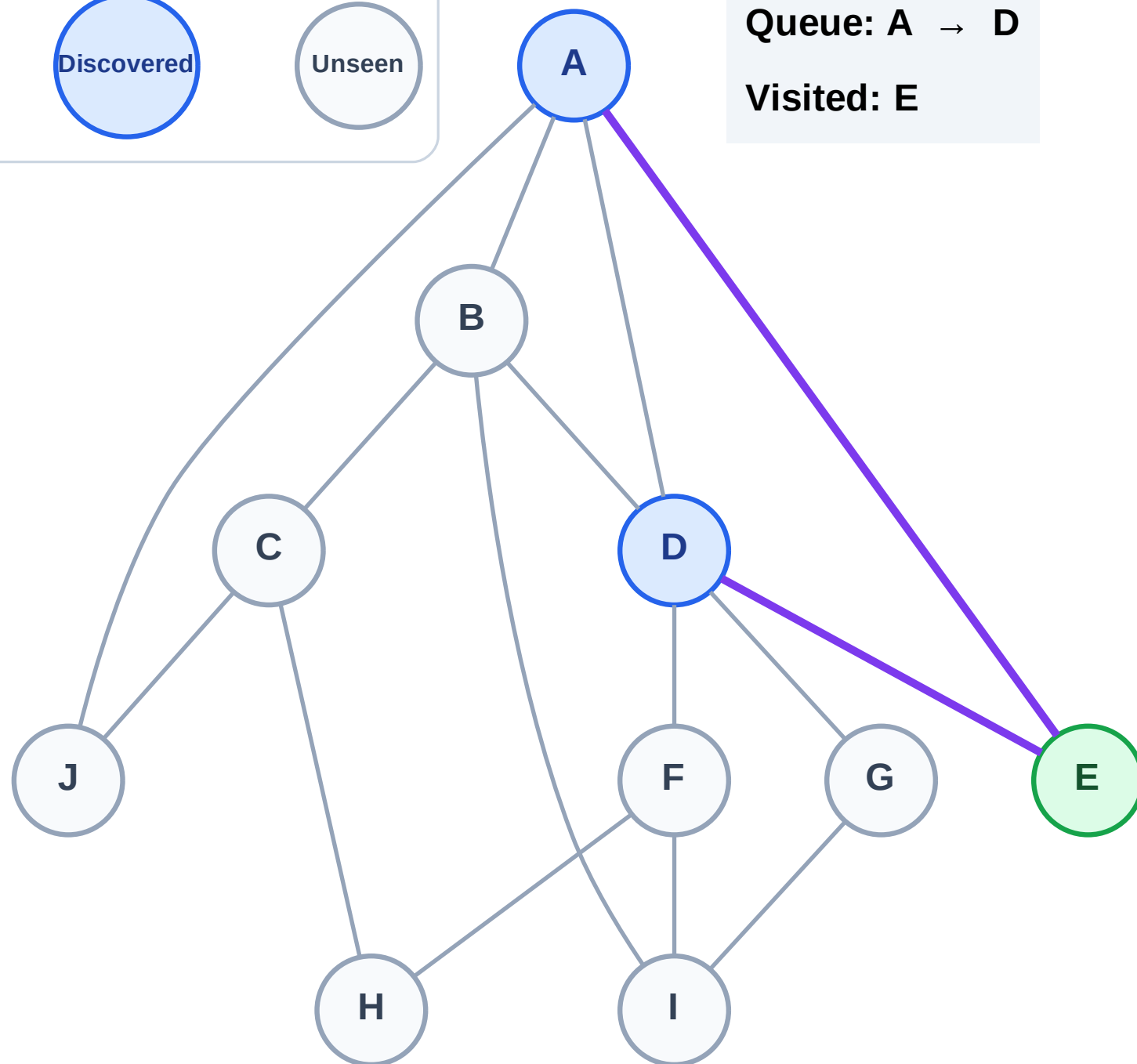
Processing

Discovered

Unseen

Queue: A → D

Visited: E



BFS Traversal

Step 4: Process node A

Legend

Complete

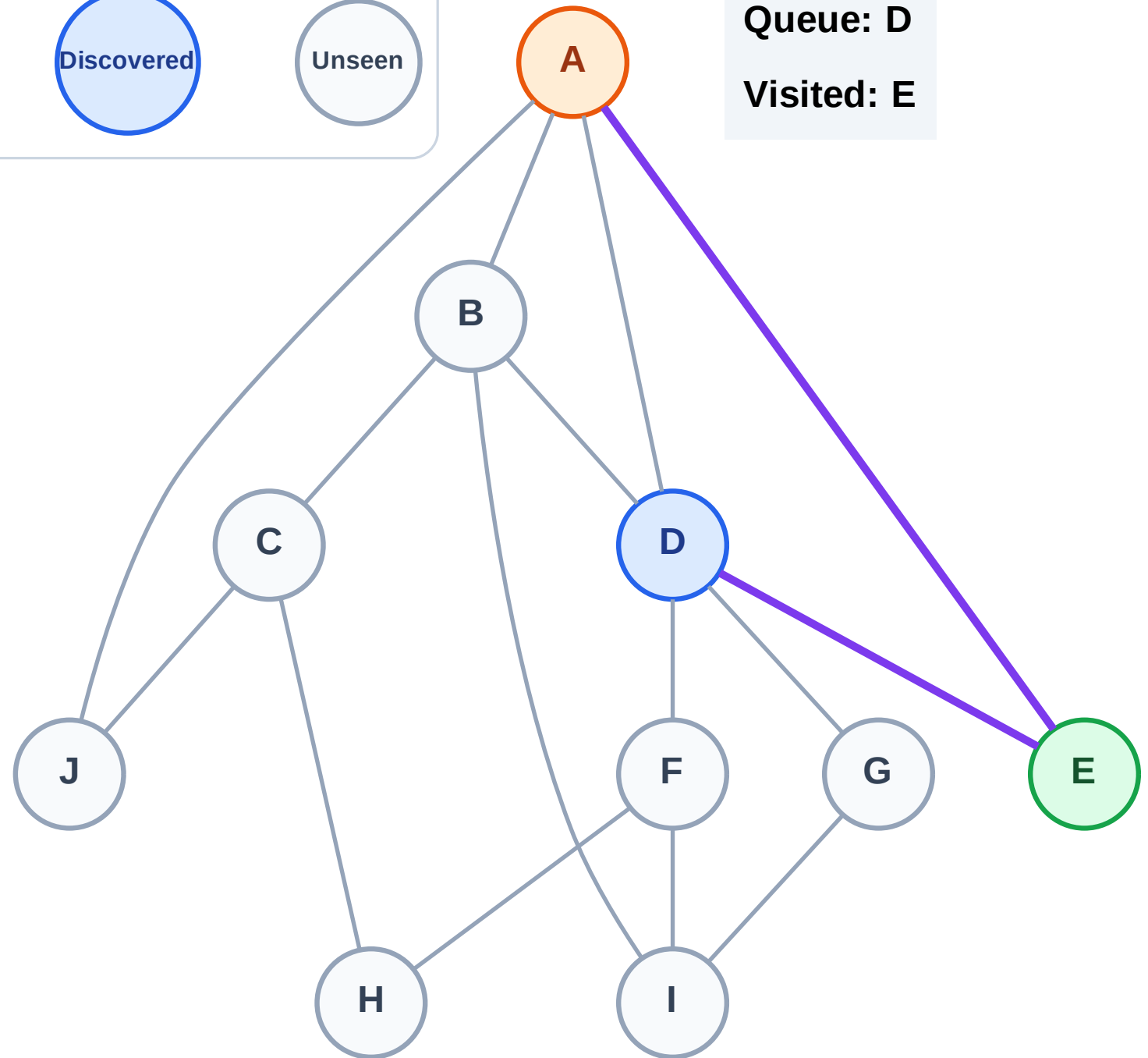
Processing

Discovered

Unseen

Queue: D

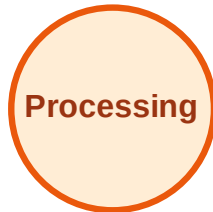
Visited: E



BFS Traversal

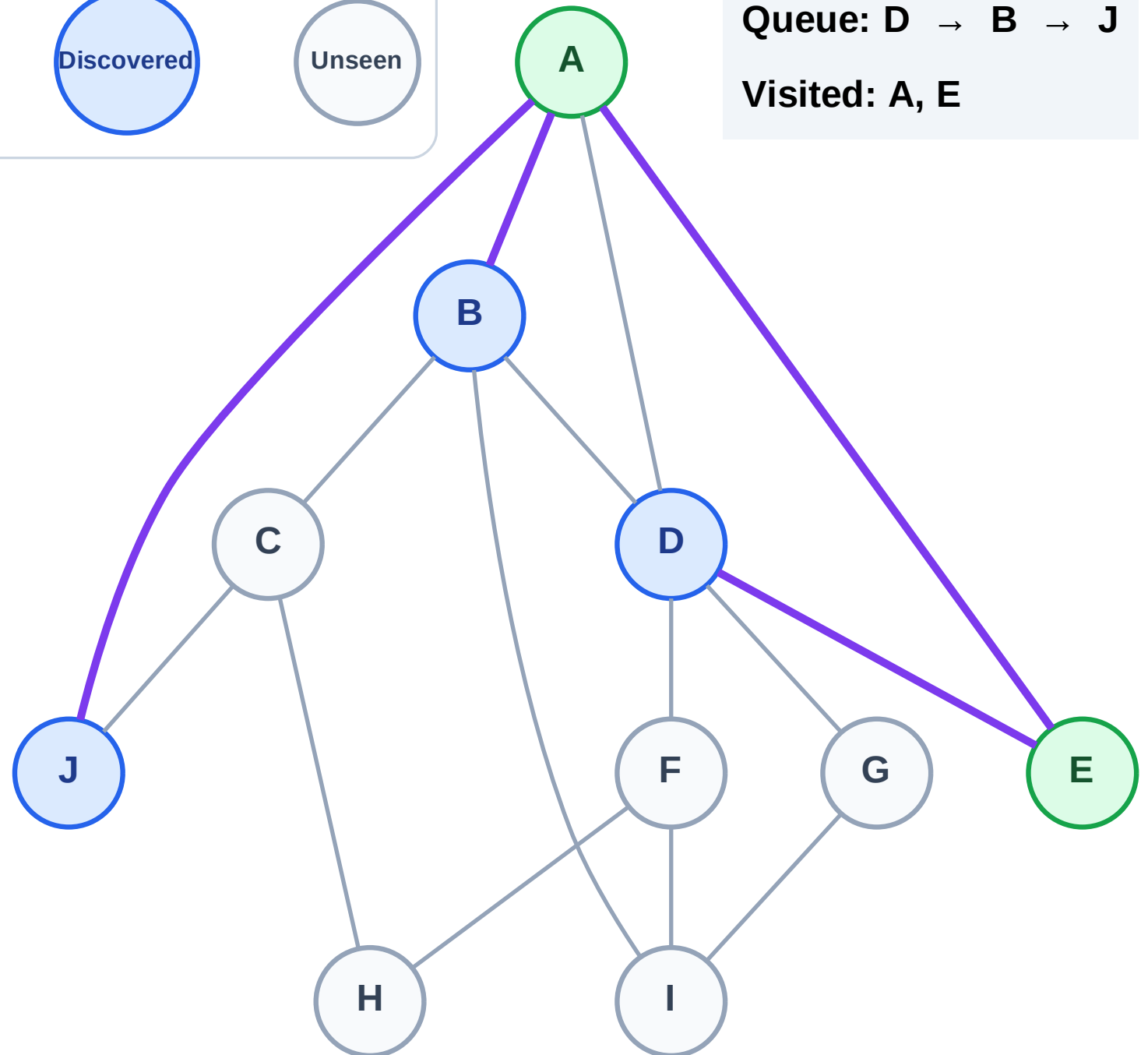
Step 5: Discover B, J from A

Legend



Queue: D → B → J

Visited: A, E



BFS Traversal

Step 6: Process node D

Legend

Complete

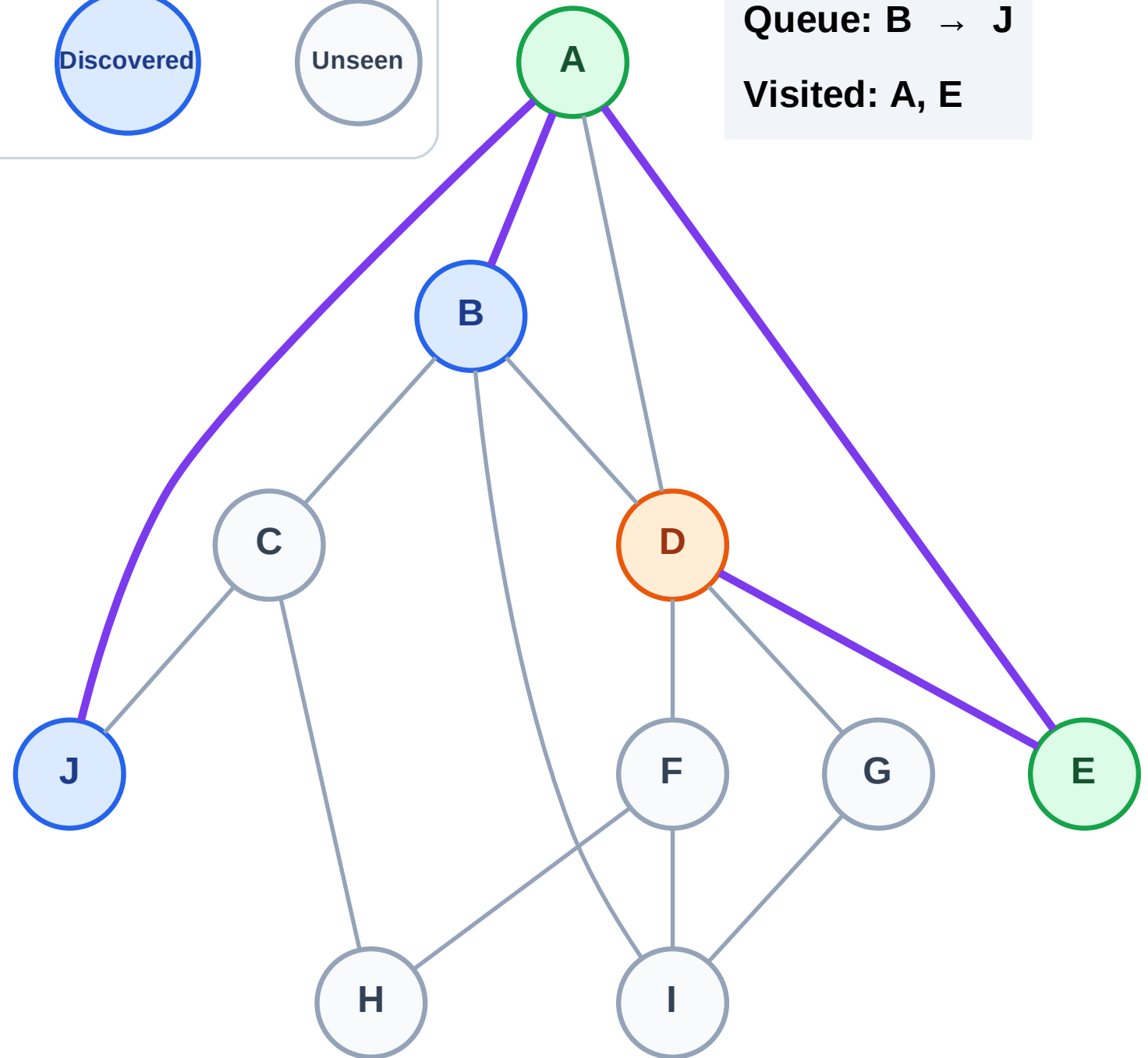
Processing

Discovered

Unseen

Queue: B → J

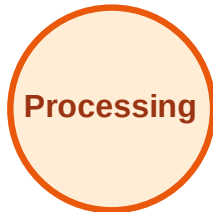
Visited: A, E



BFS Traversal

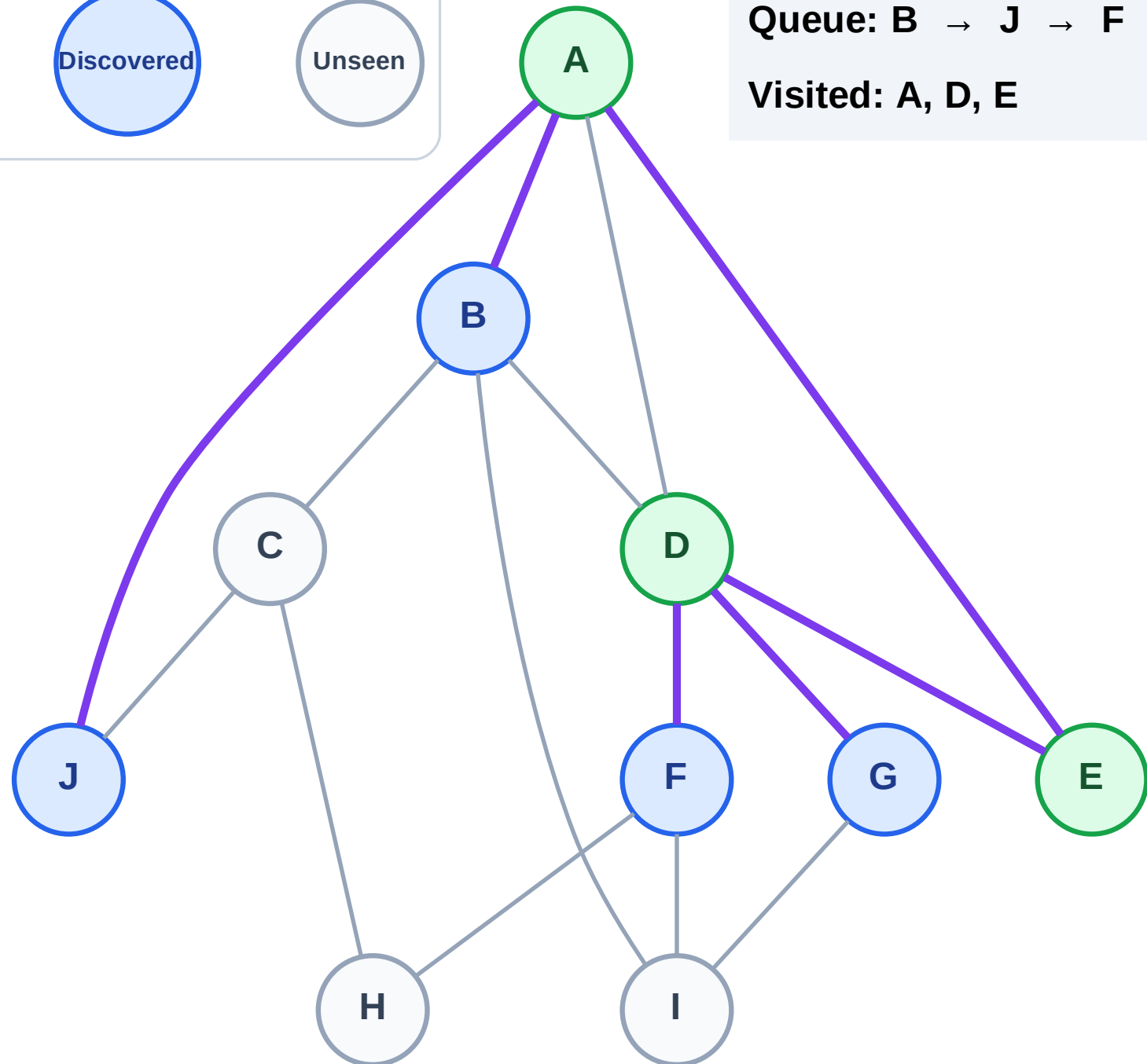
Step 7: Discover F, G from D

Legend



Queue: B → J → F → G

Visited: A, D, E



BFS Traversal

Step 8: Process node B

Legend

Complete

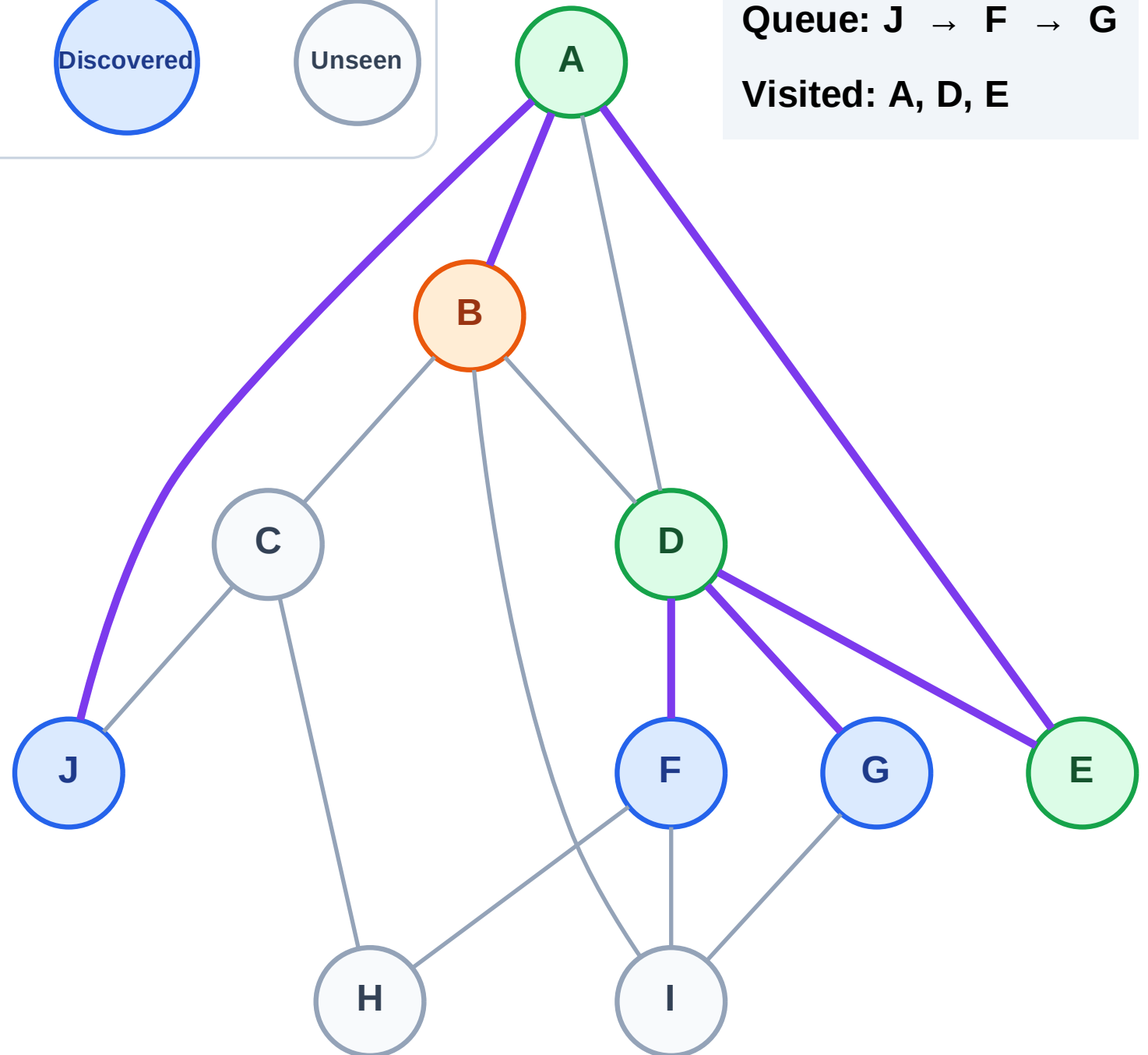
Processing

Discovered

Unseen

Queue: J → F → G

Visited: A, D, E



BFS Traversal

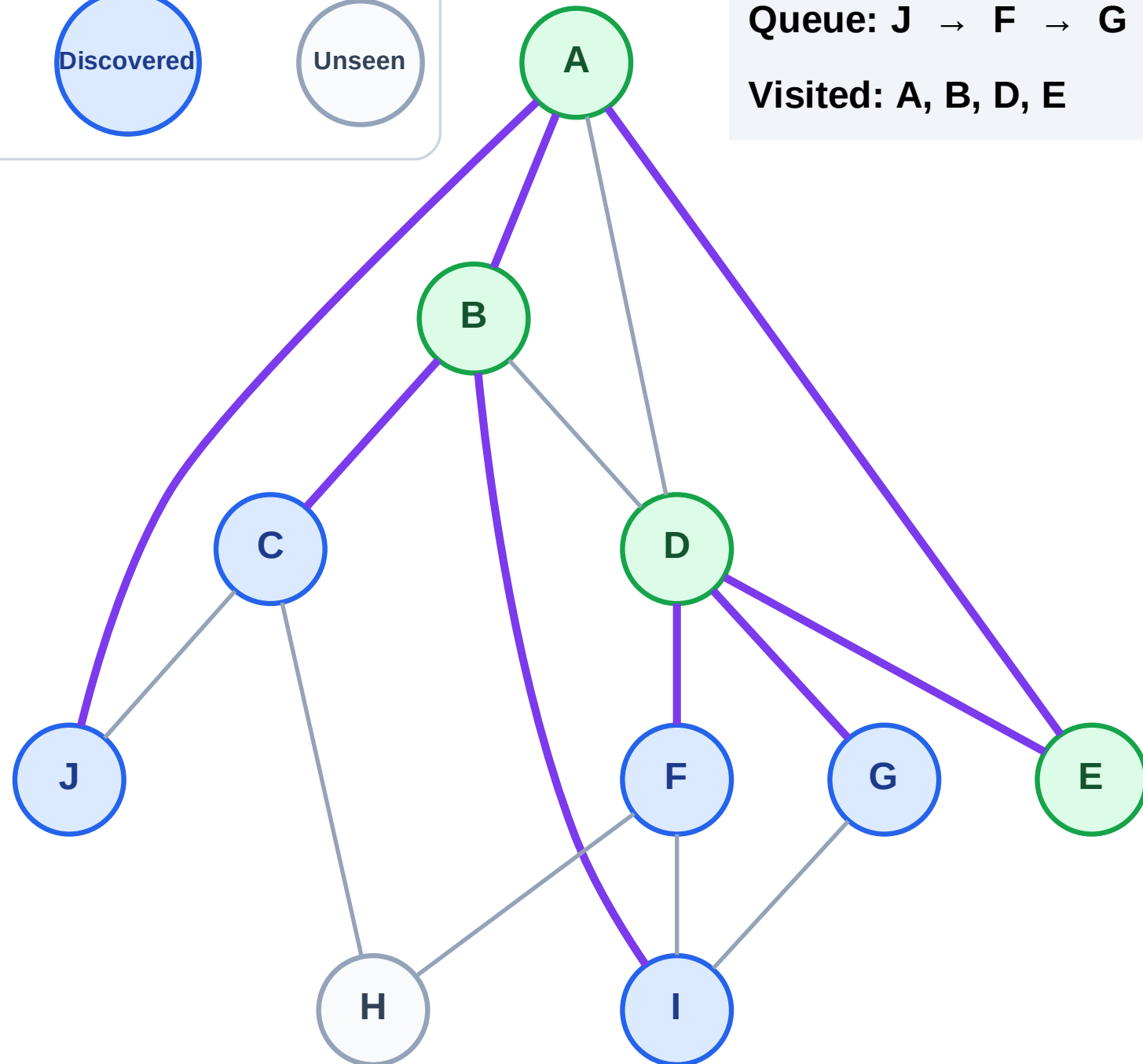
Step 9: Discover C, I from B

Legend



Queue: J → F → G → C → I

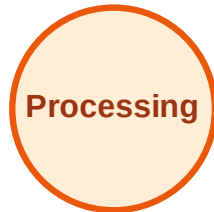
Visited: A, B, D, E



BFS Traversal

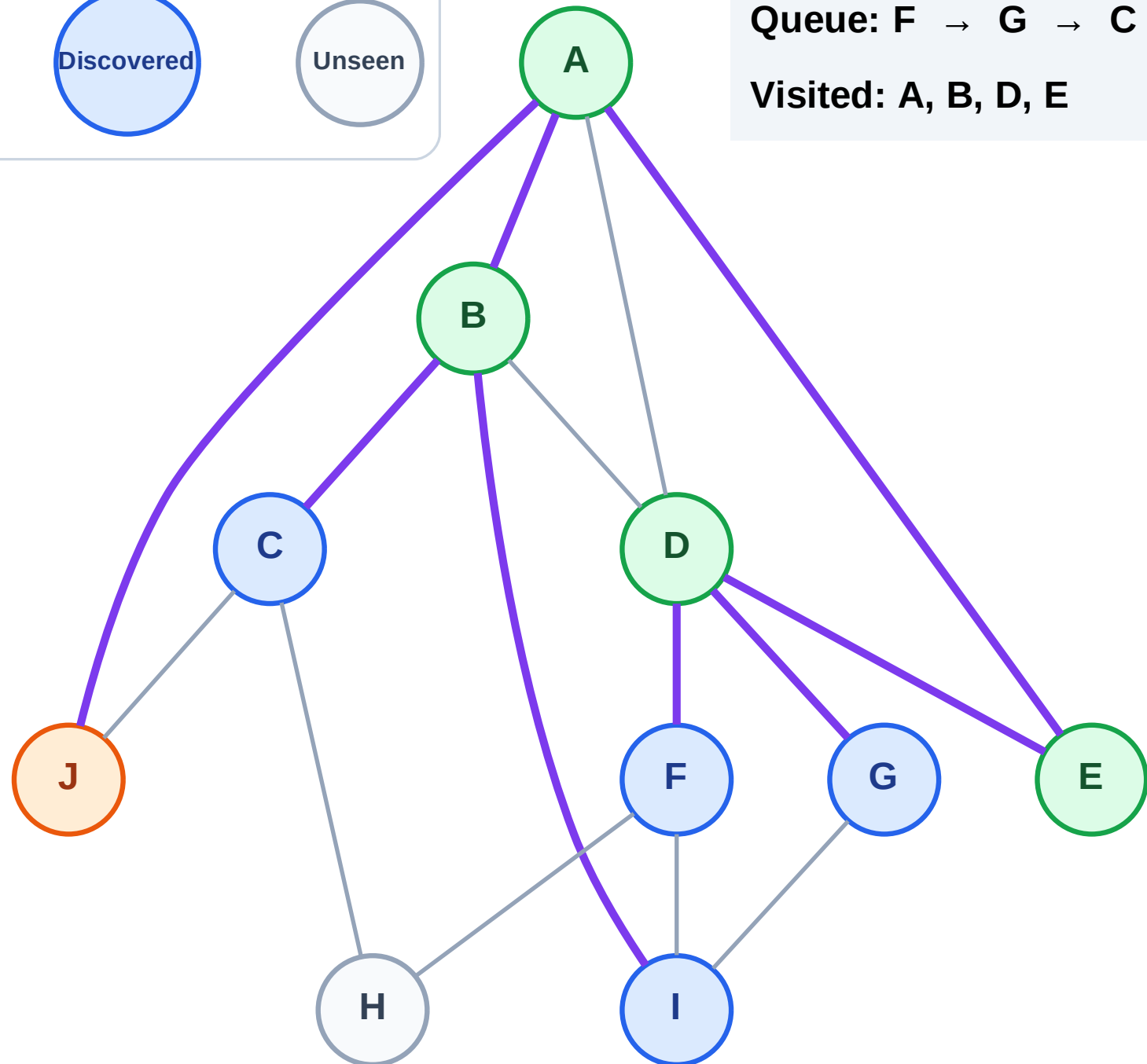
Step 10: Process node J

Legend



Queue: F → G → C → I

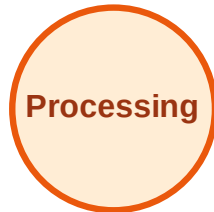
Visited: A, B, D, E



BFS Traversal

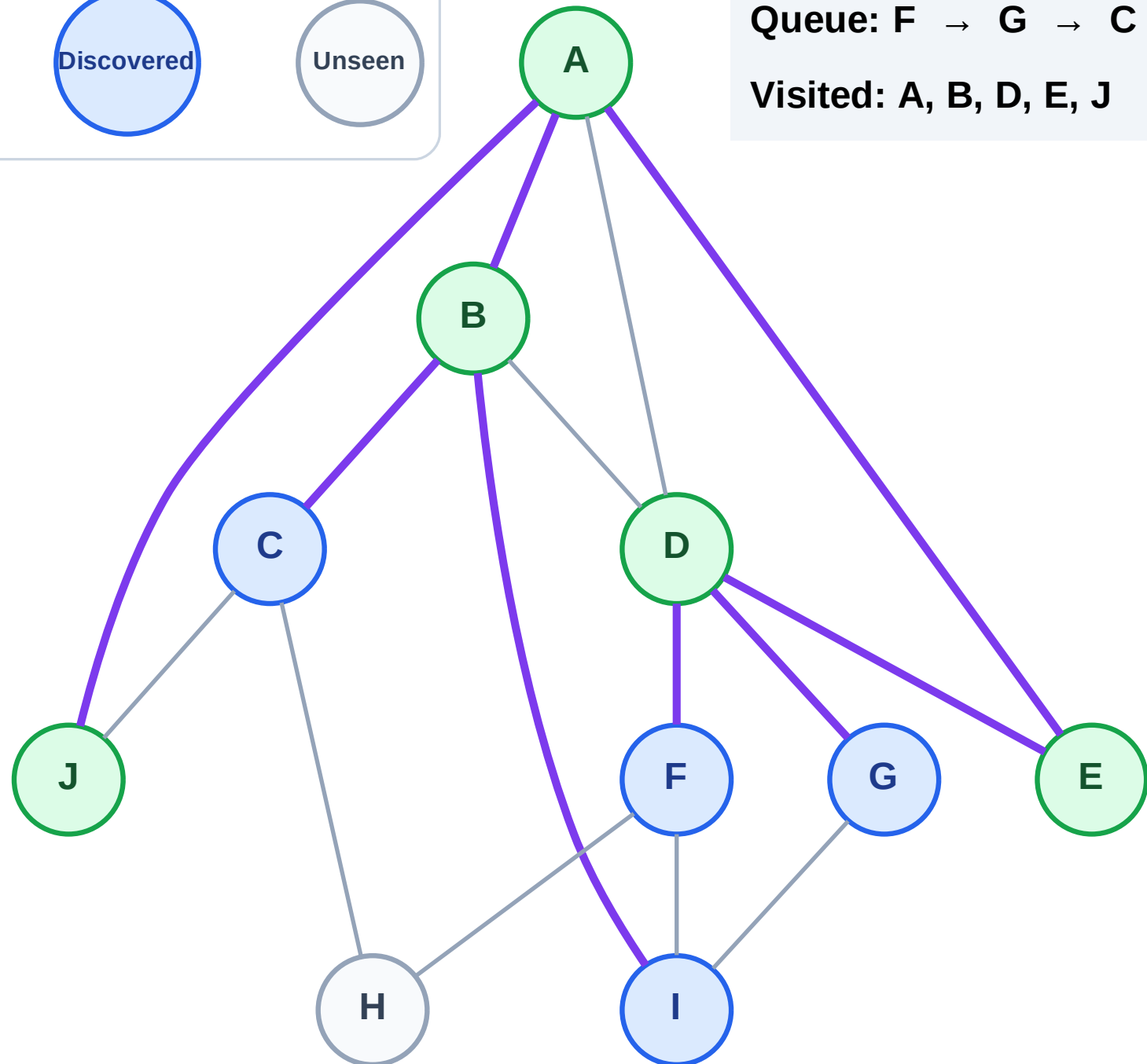
Step 11: No new neighbors from J

Legend



Queue: F → G → C → I

Visited: A, B, D, E, J



BFS Traversal

Step 12: Process node F

Legend

Complete

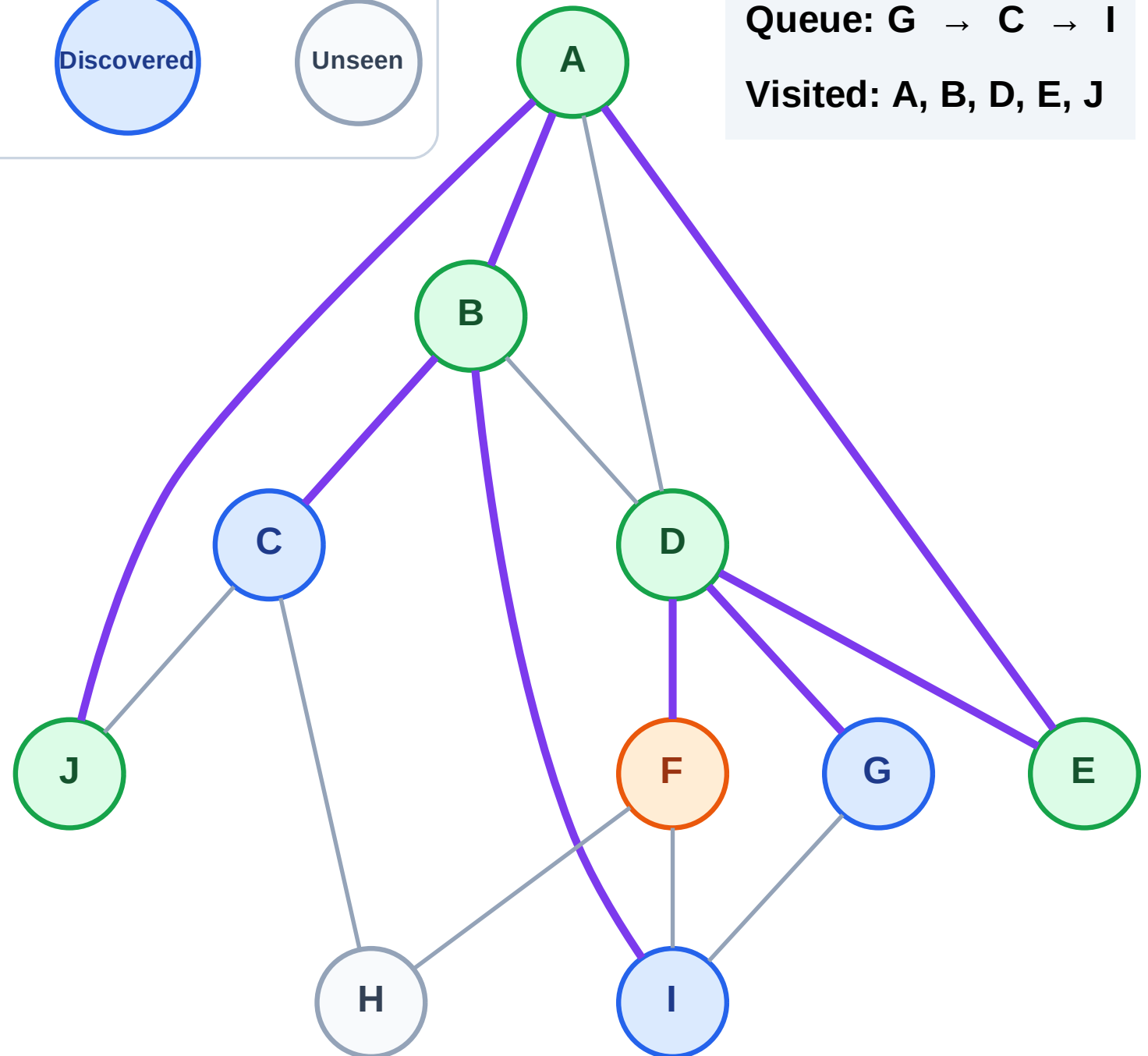
Processing

Discovered

Unseen

Queue: G → C → I

Visited: A, B, D, E, J



BFS Traversal

Step 13: Discover H from F

Legend

Complete

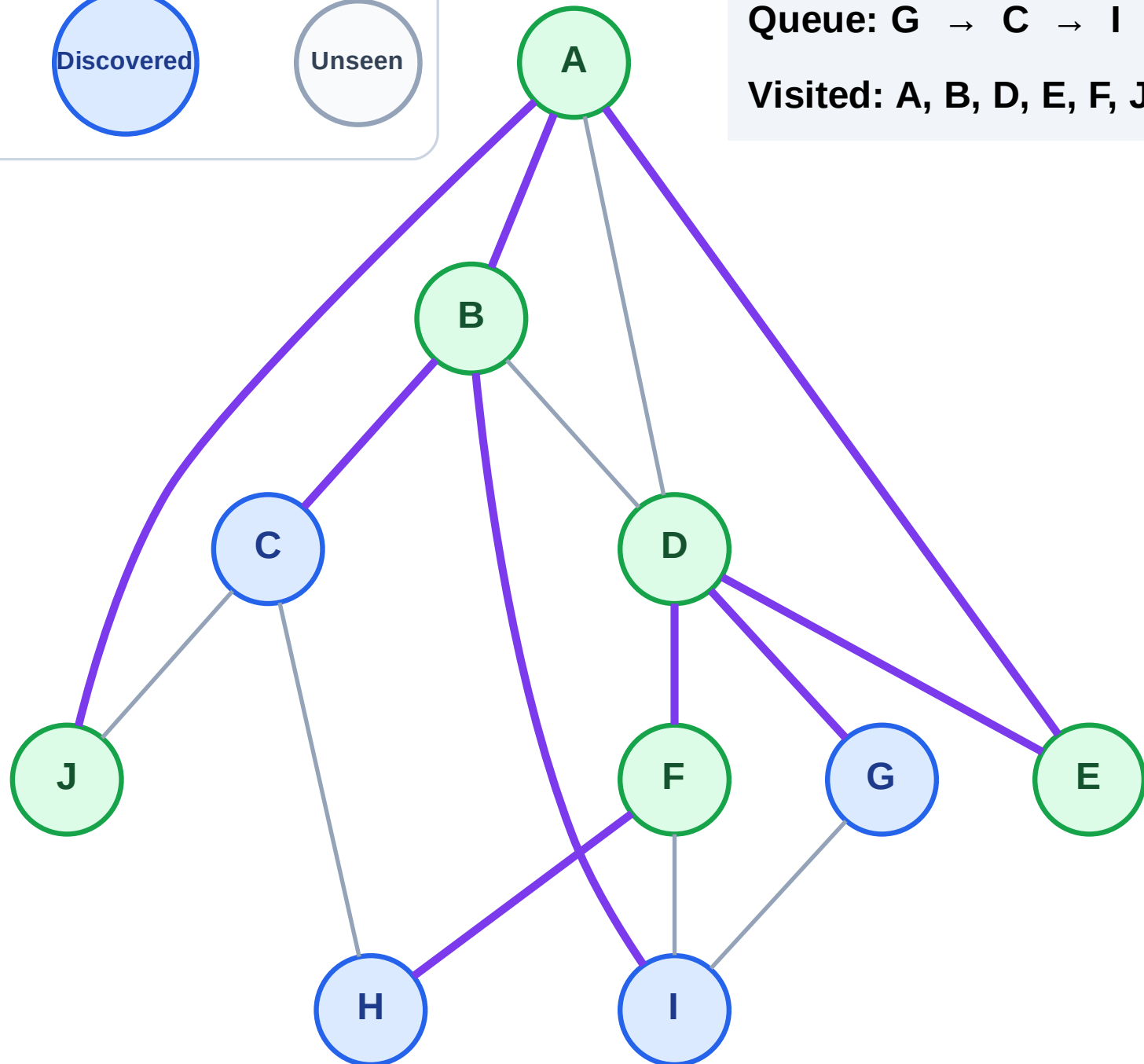
Processing

Discovered

Unseen

Queue: G → C → I → H

Visited: A, B, D, E, F, J



BFS Traversal

Step 14: Process node G

Legend

Complete

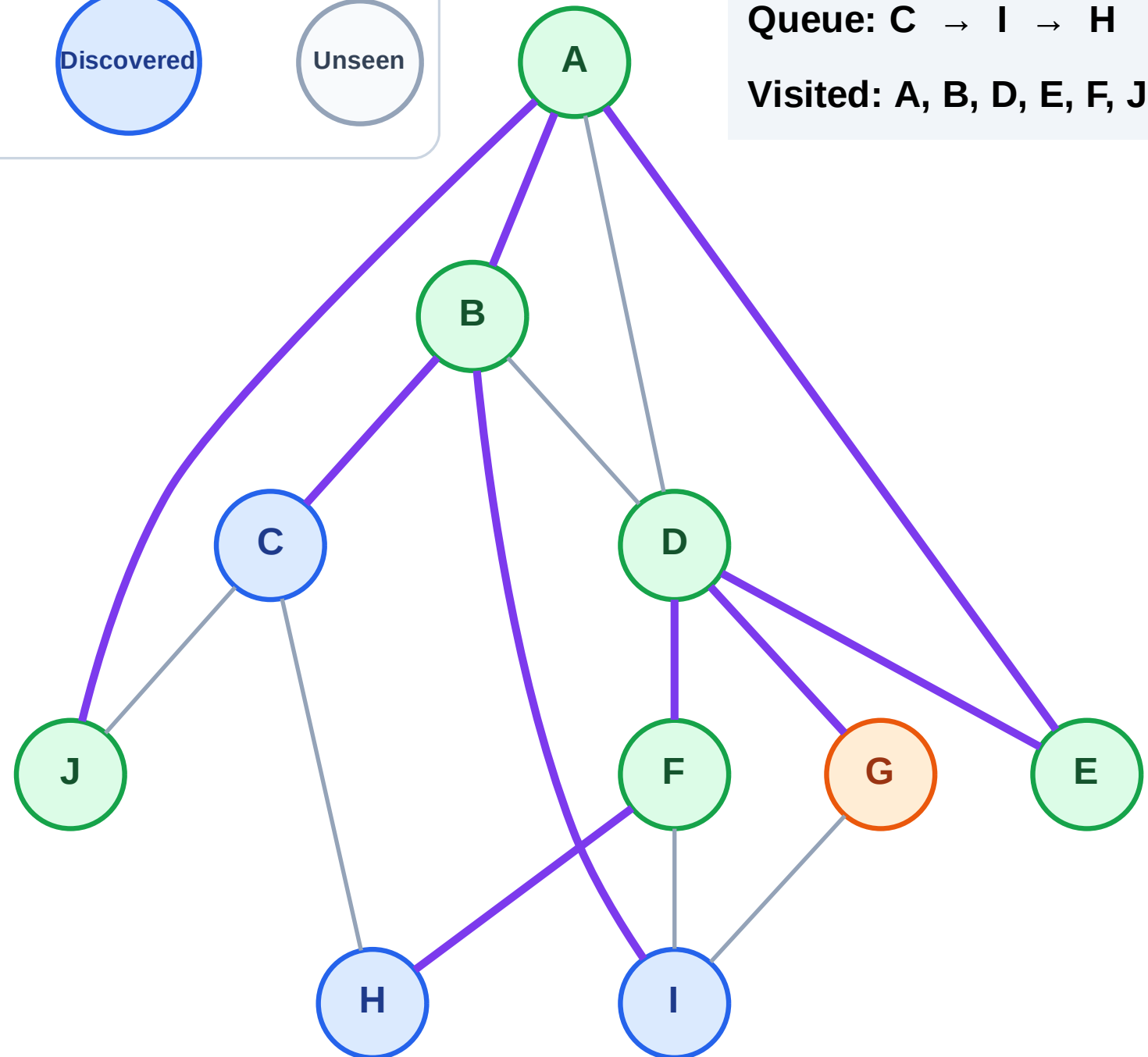
Processing

Discovered

Unseen

Queue: C → I → H

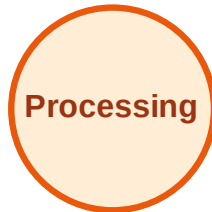
Visited: A, B, D, E, F, J



BFS Traversal

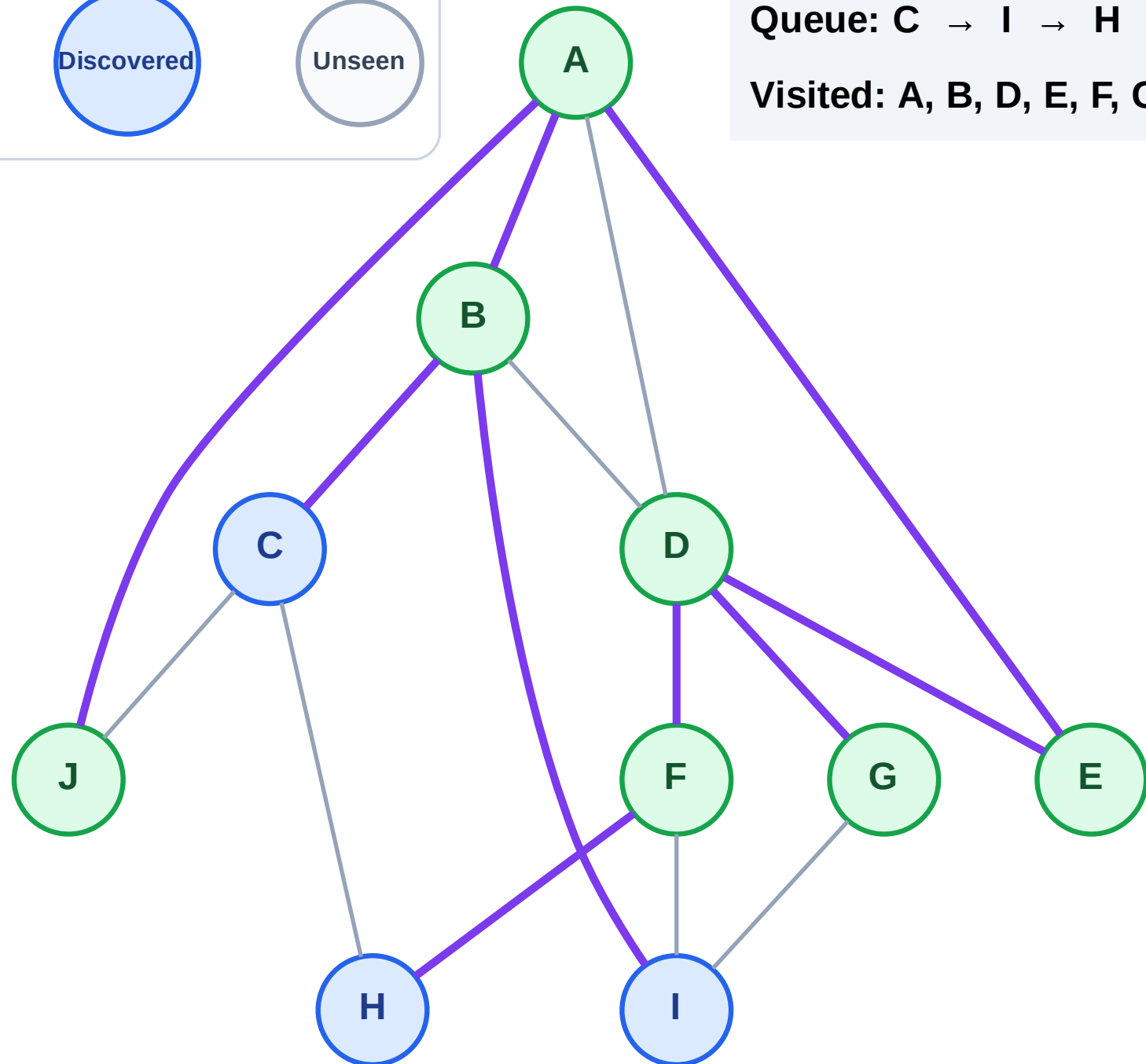
Step 15: No new neighbors from G

Legend



Queue: C → I → H

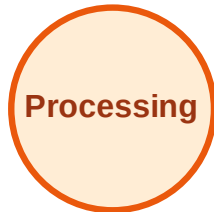
Visited: A, B, D, E, F, G, J



BFS Traversal

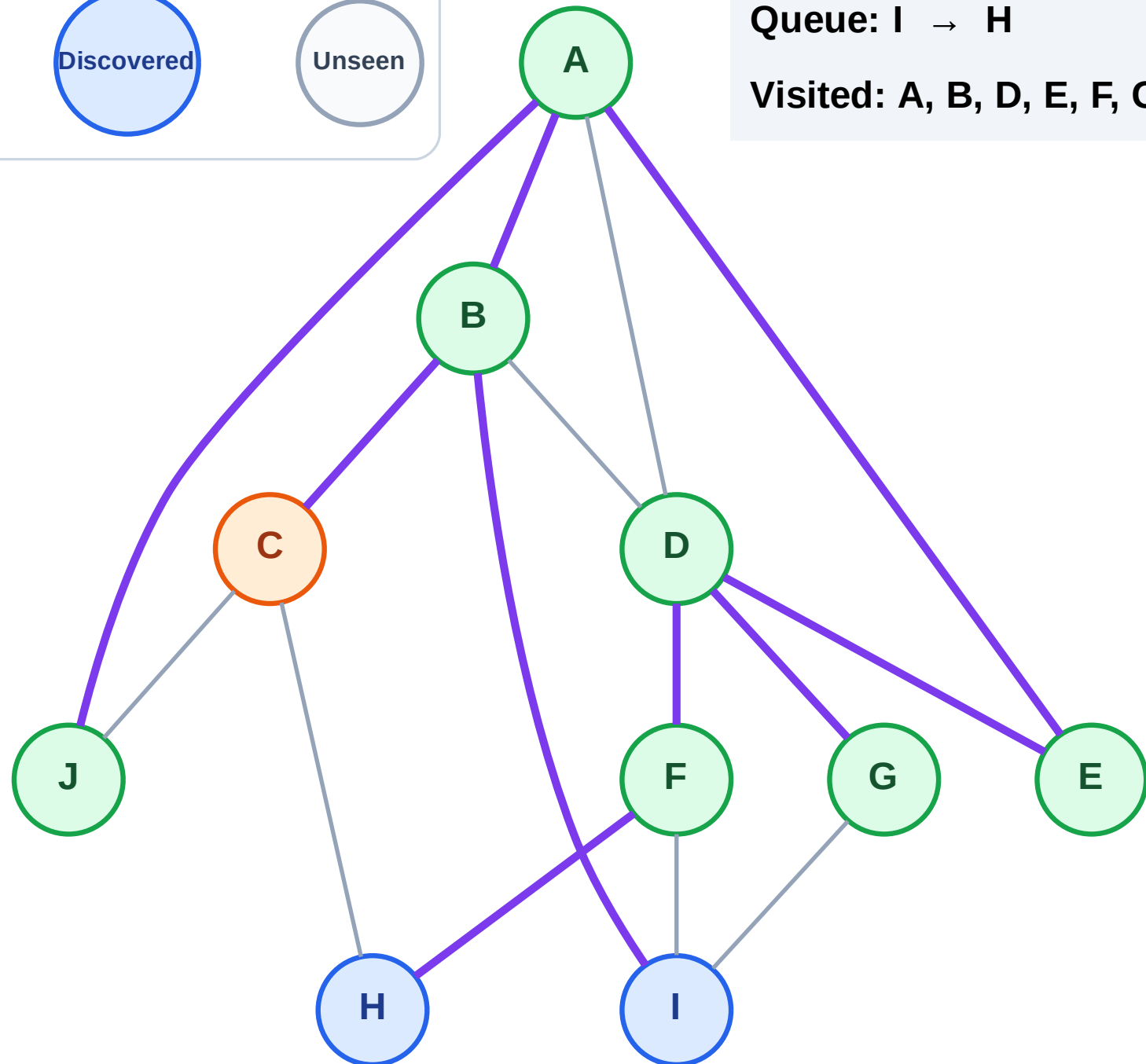
Step 16: Process node C

Legend



Queue: I → H

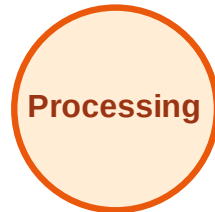
Visited: A, B, D, E, F, G, J



BFS Traversal

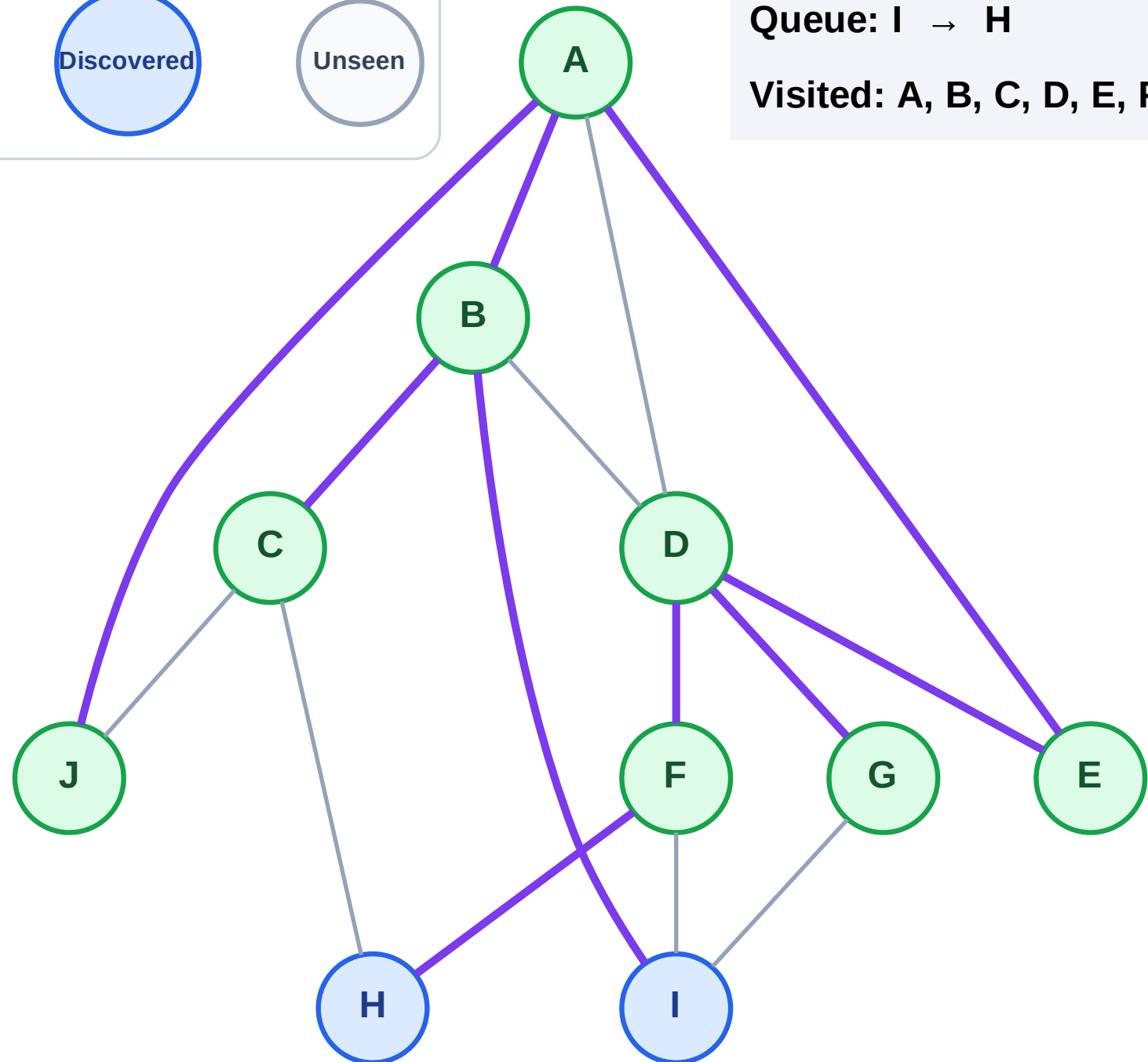
Step 17: No new neighbors from C

Legend



Queue: I → H

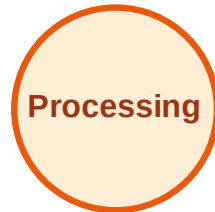
Visited: A, B, C, D, E, F, G, J



BFS Traversal

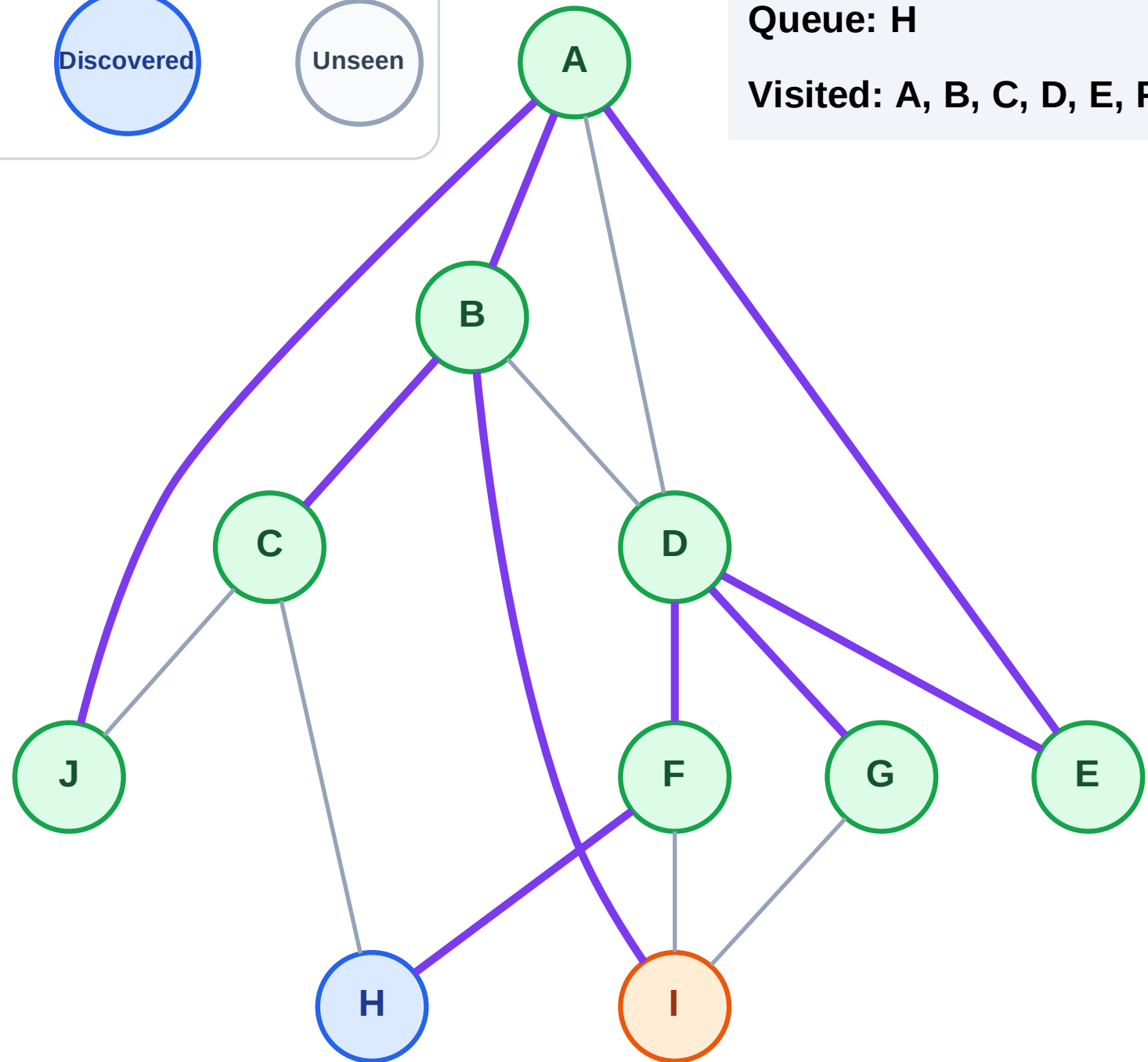
Step 18: Process node I

Legend



Queue: H

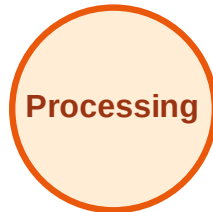
Visited: A, B, C, D, E, F, G, J



BFS Traversal

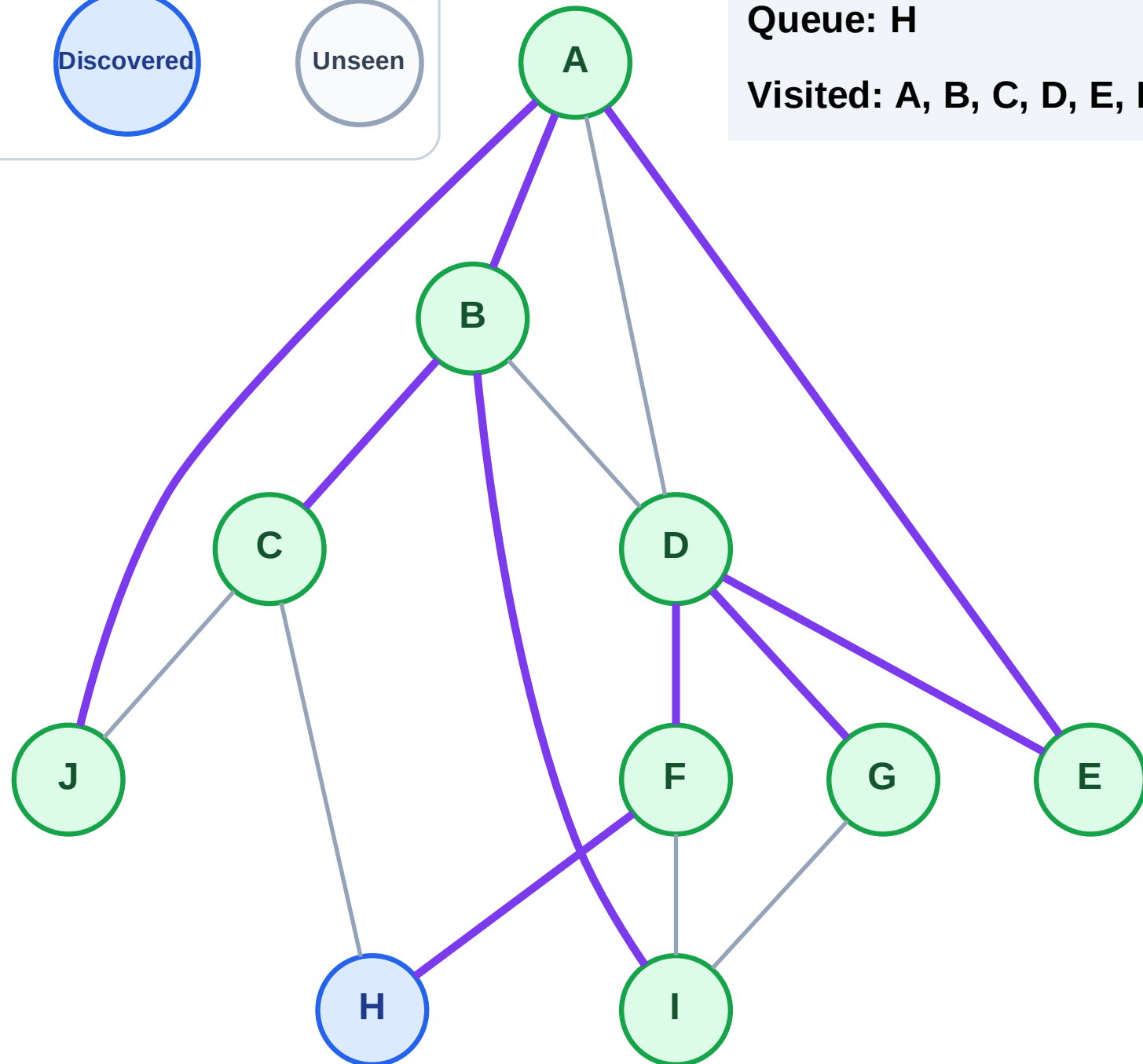
Step 19: No new neighbors from I

Legend



Queue: H

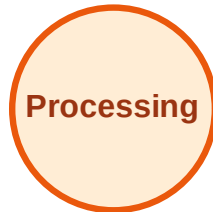
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

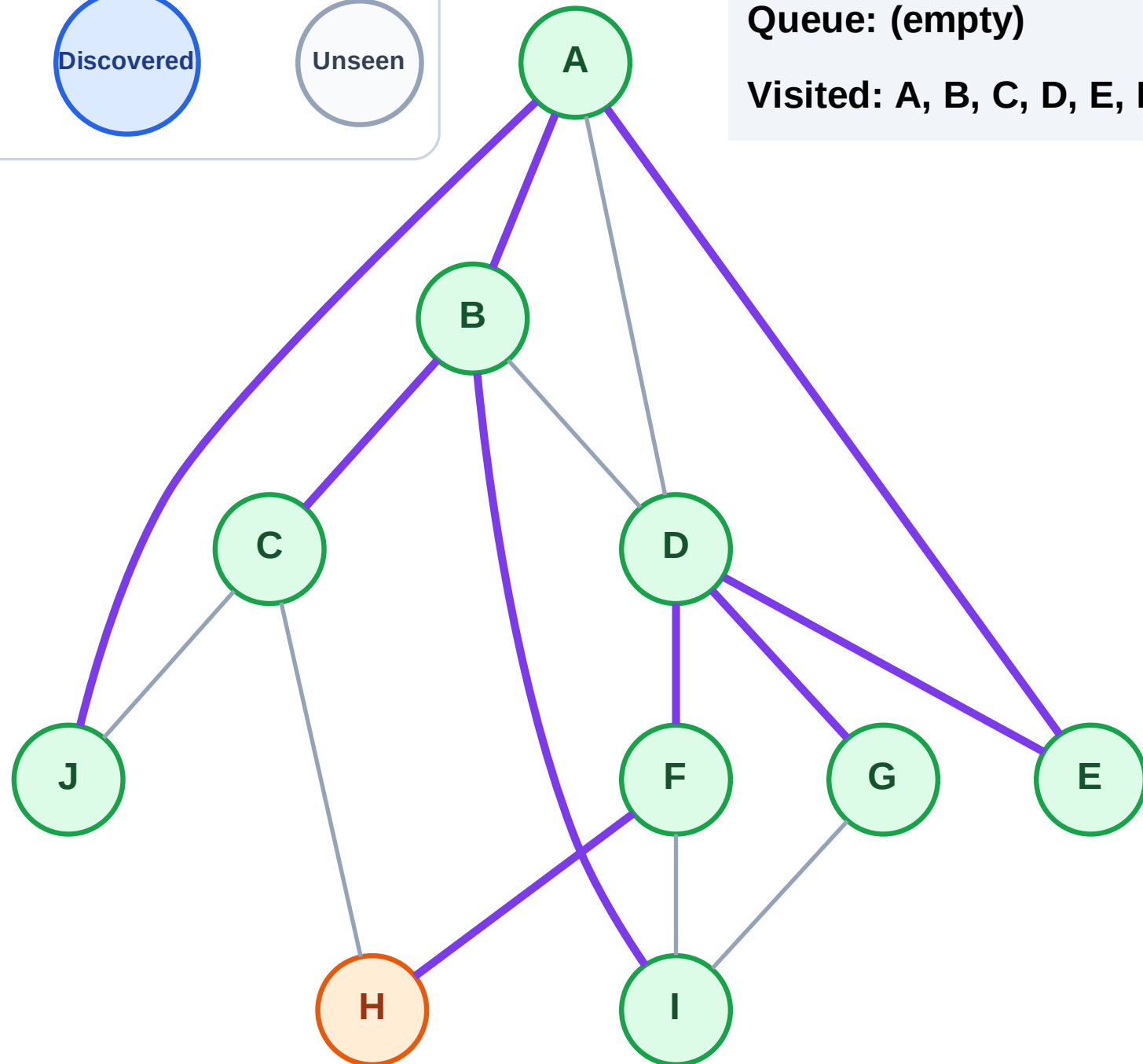
Step 20: Process node H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

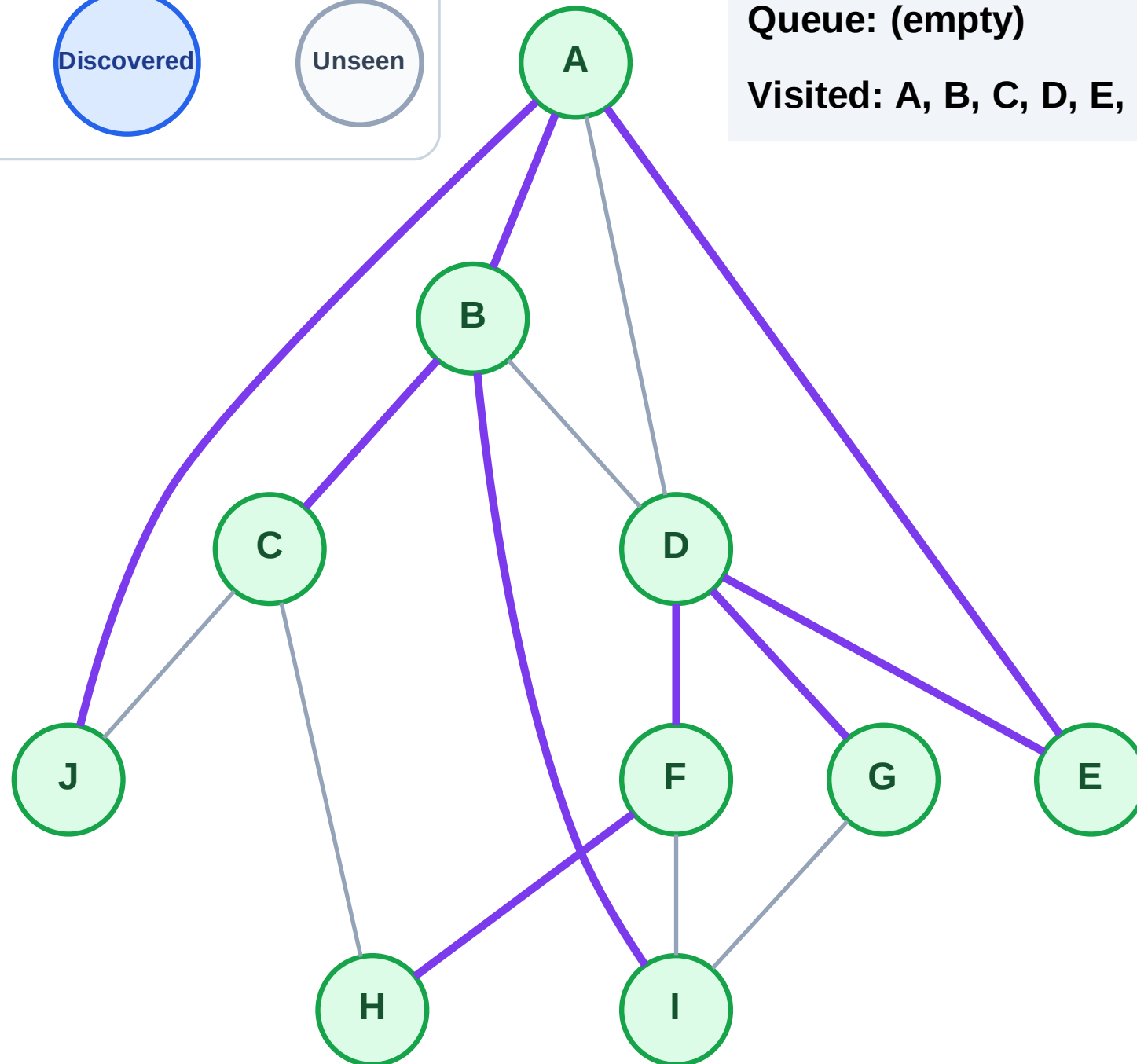
Step 21: No new neighbors from H

Legend



Queue: (empty)

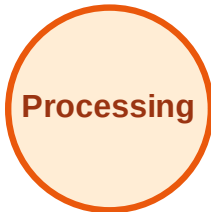
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

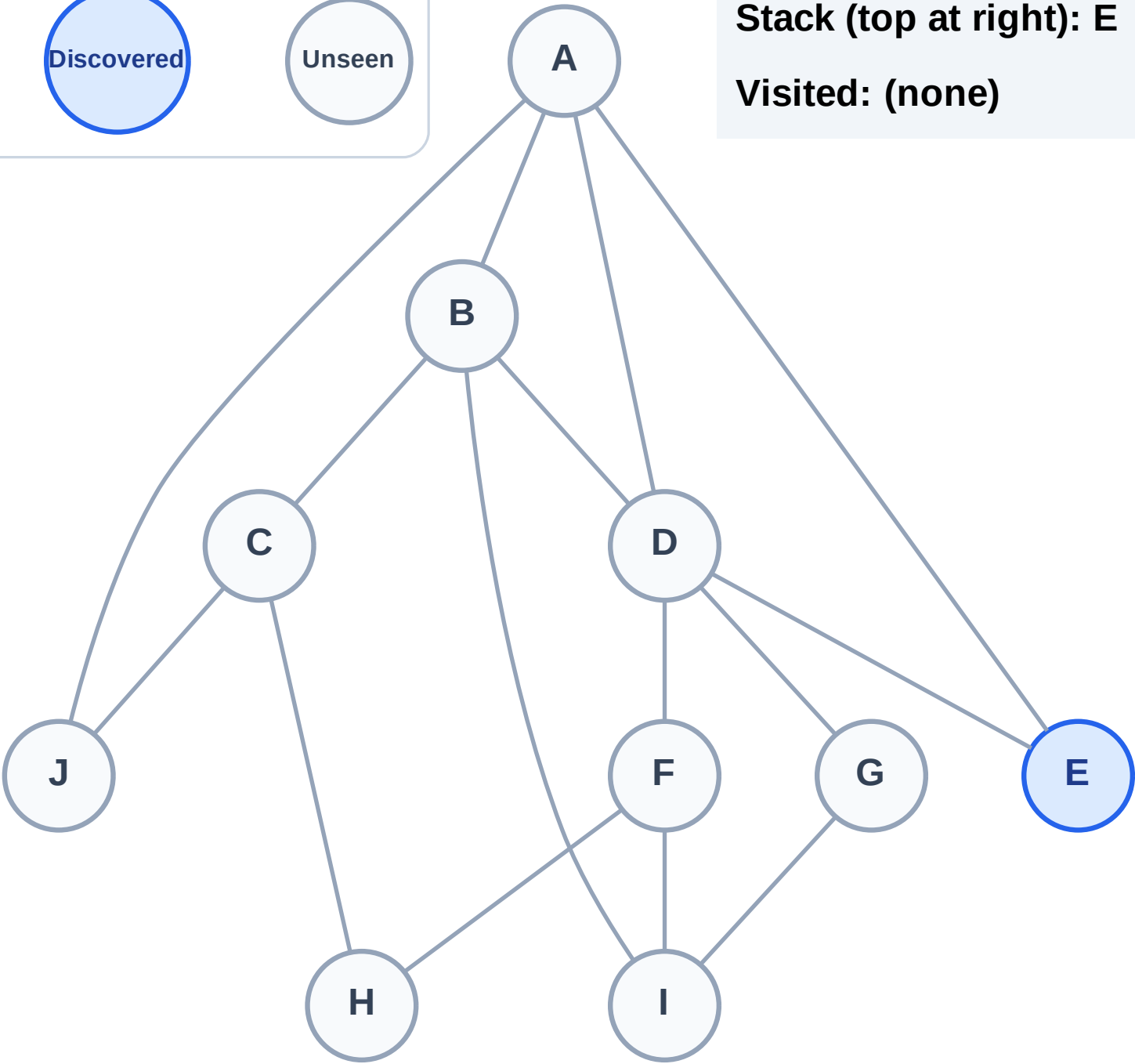
Step 1: Start at E

Legend



Stack (top at right): E

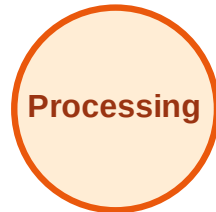
Visited: (none)



DFS Traversal

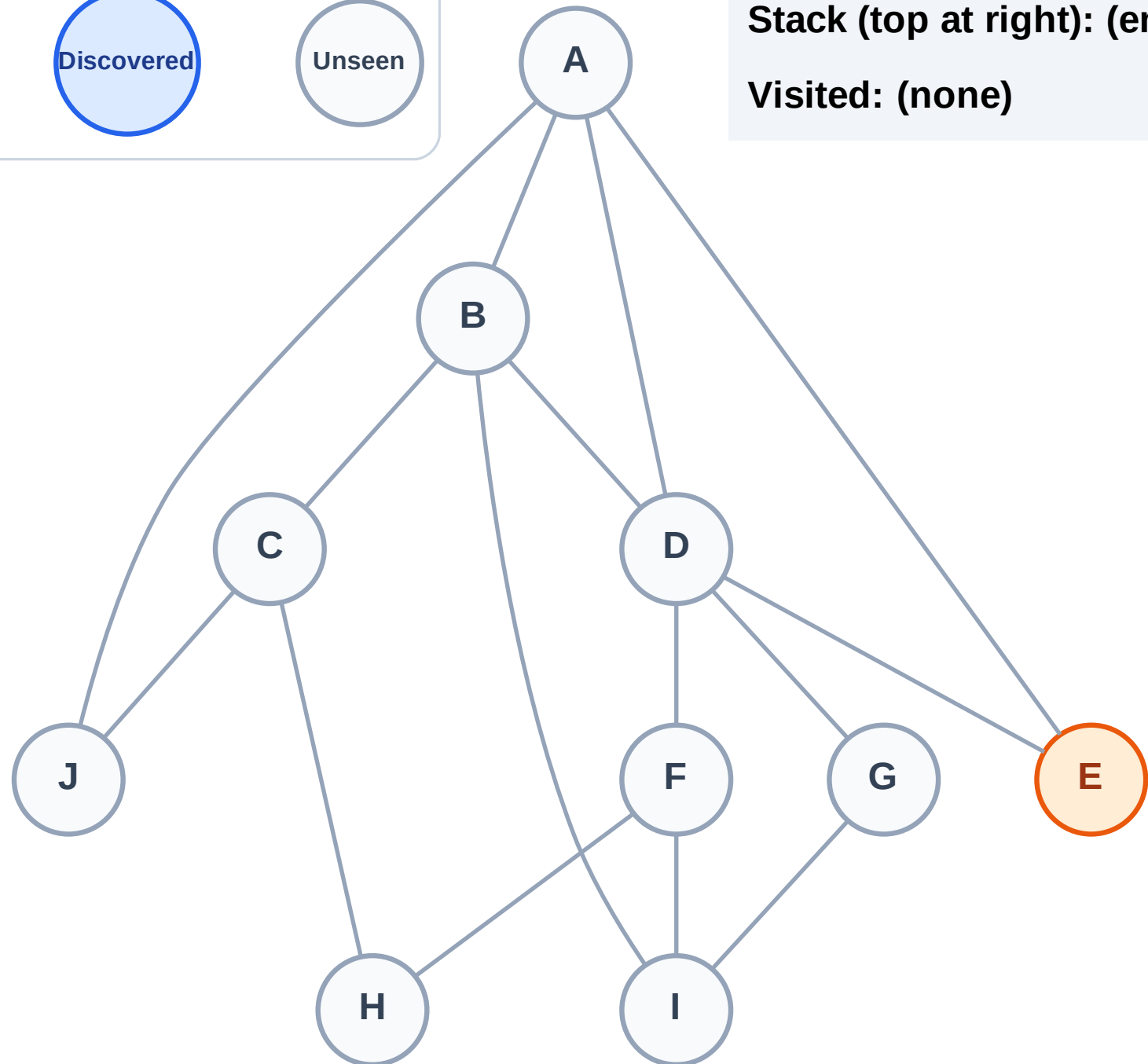
Step 2: Process node E

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover D, A from E

Legend

Complete

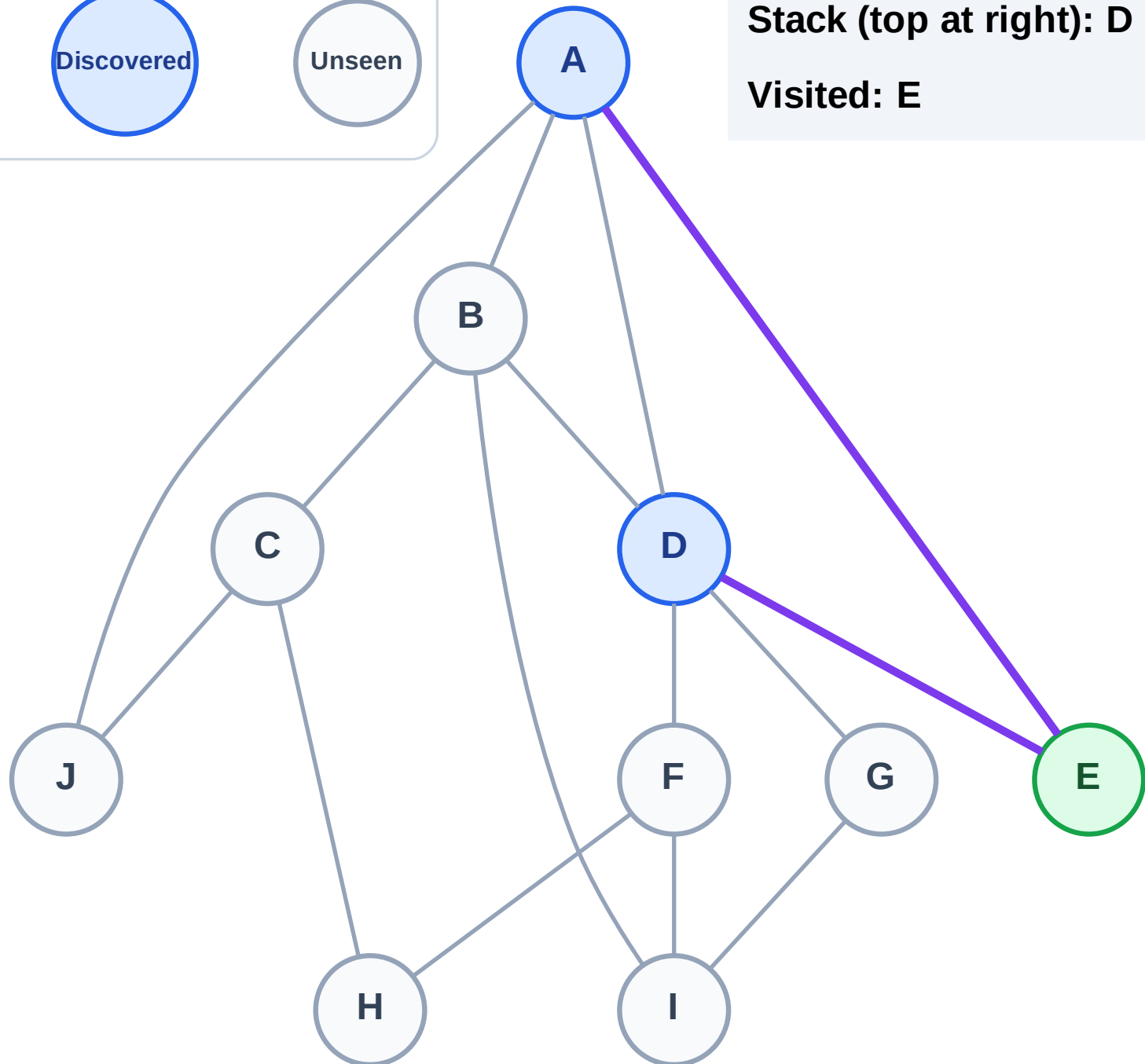
Processing

Discovered

Unseen

Stack (top at right): D | A

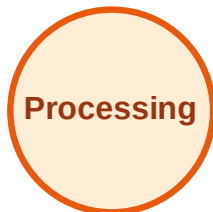
Visited: E



DFS Traversal

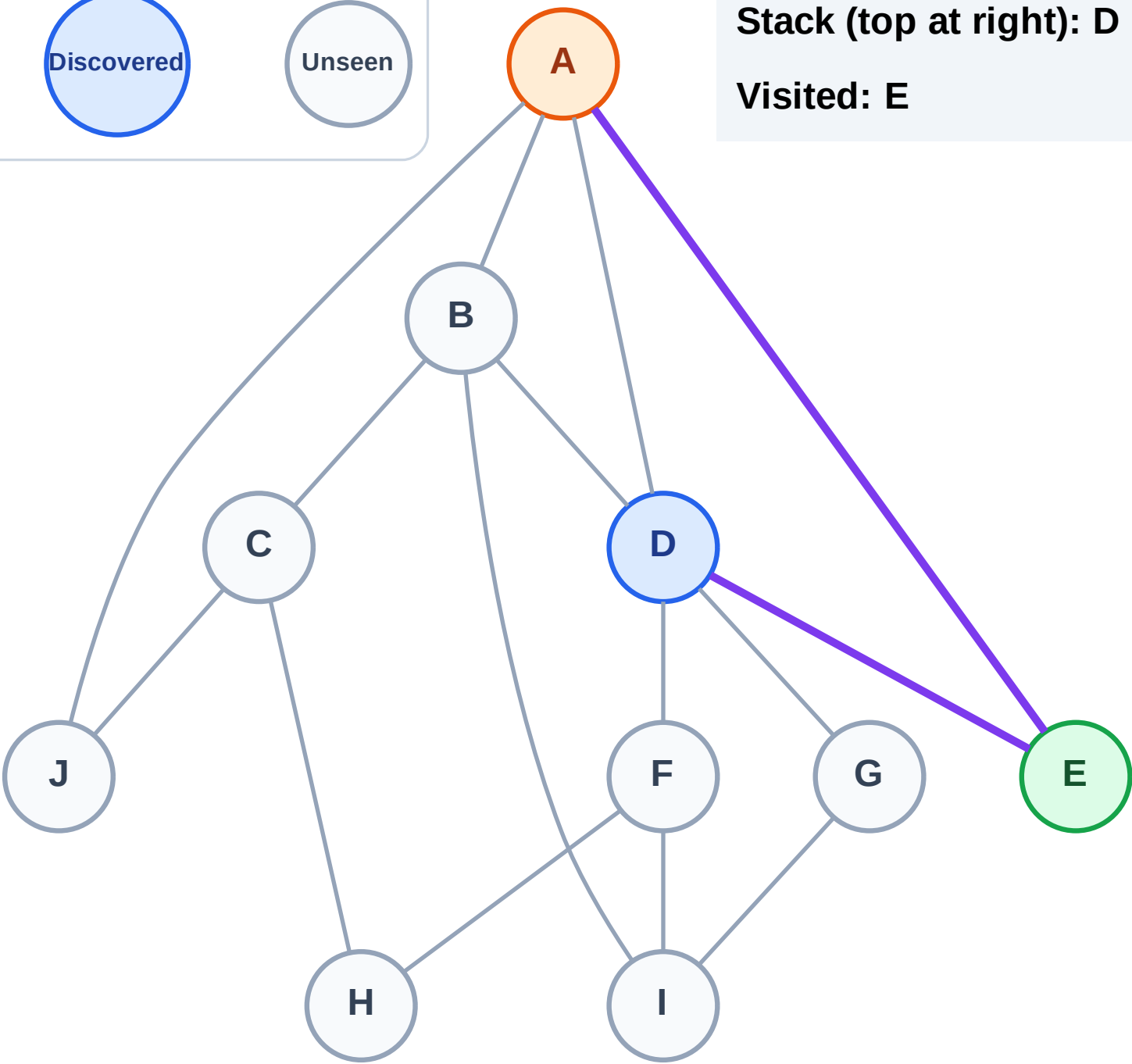
Step 4: Process node A

Legend



Stack (top at right): D

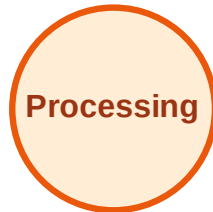
Visited: E



DFS Traversal

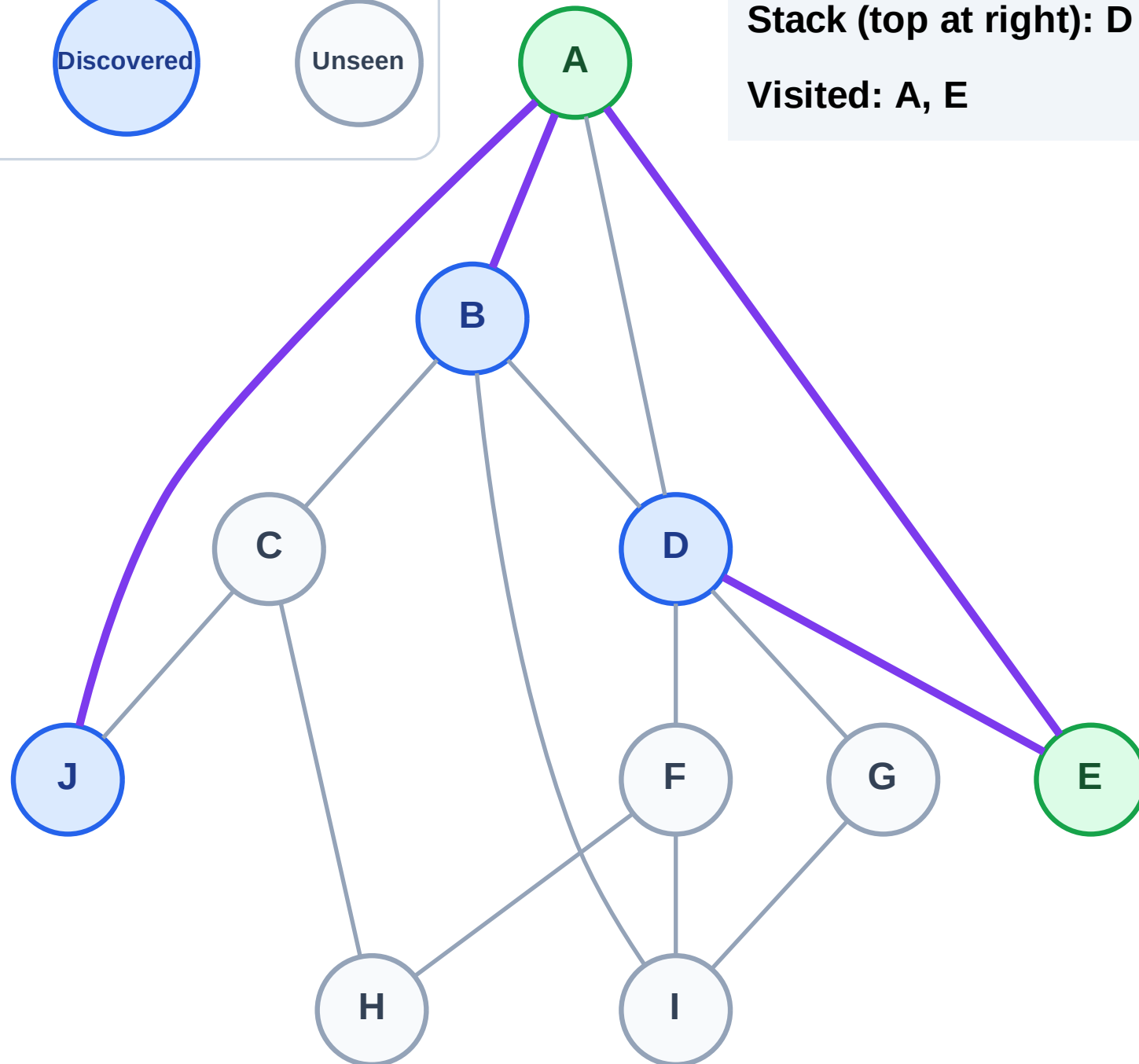
Step 5: Discover J, B from A

Legend



Stack (top at right): D | J | B

Visited: A, E



DFS Traversal

Step 6: Process node B

Legend

Complete

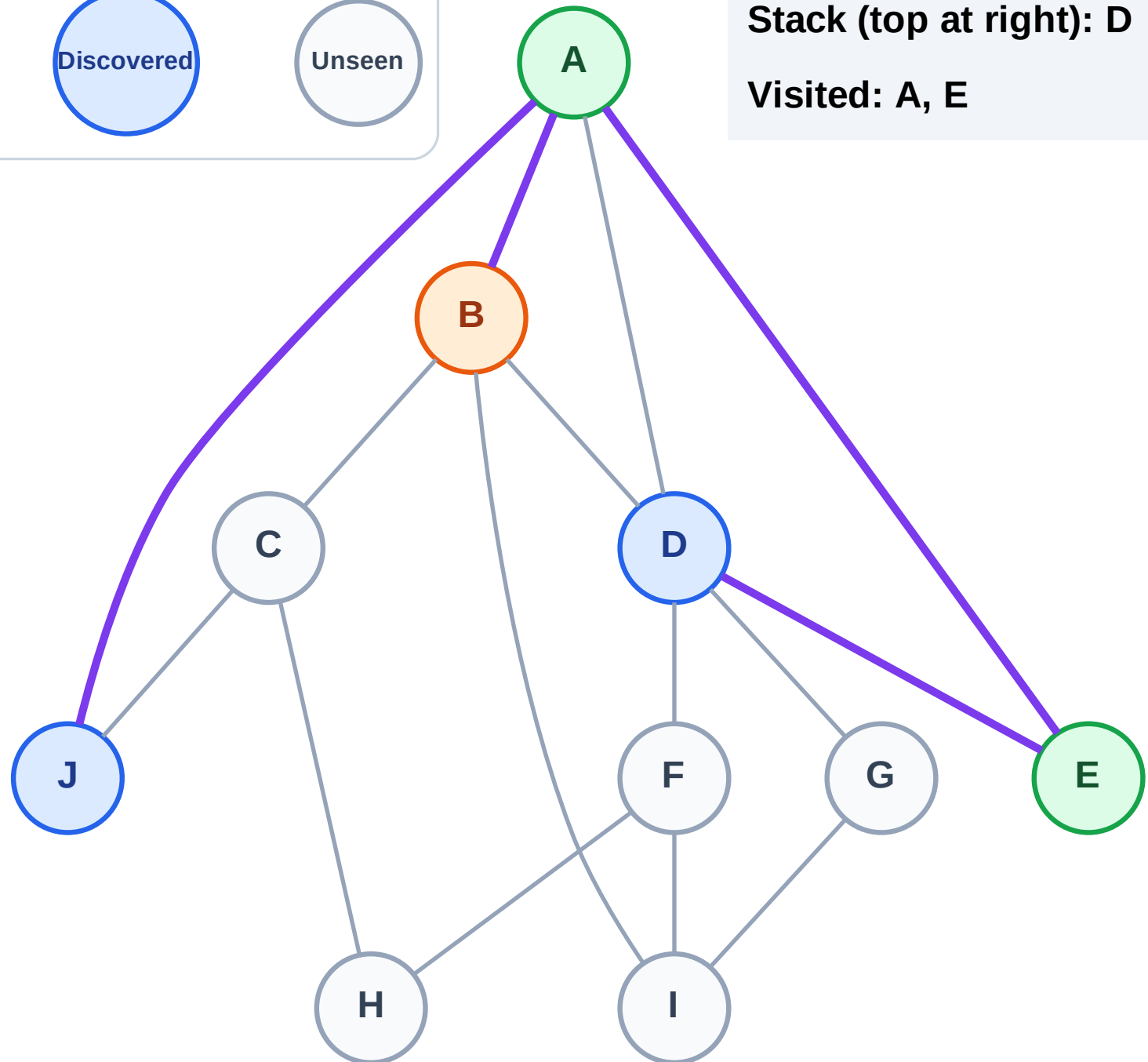
Processing

Discovered

Unseen

Stack (top at right): D | J

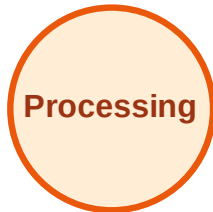
Visited: A, E



DFS Traversal

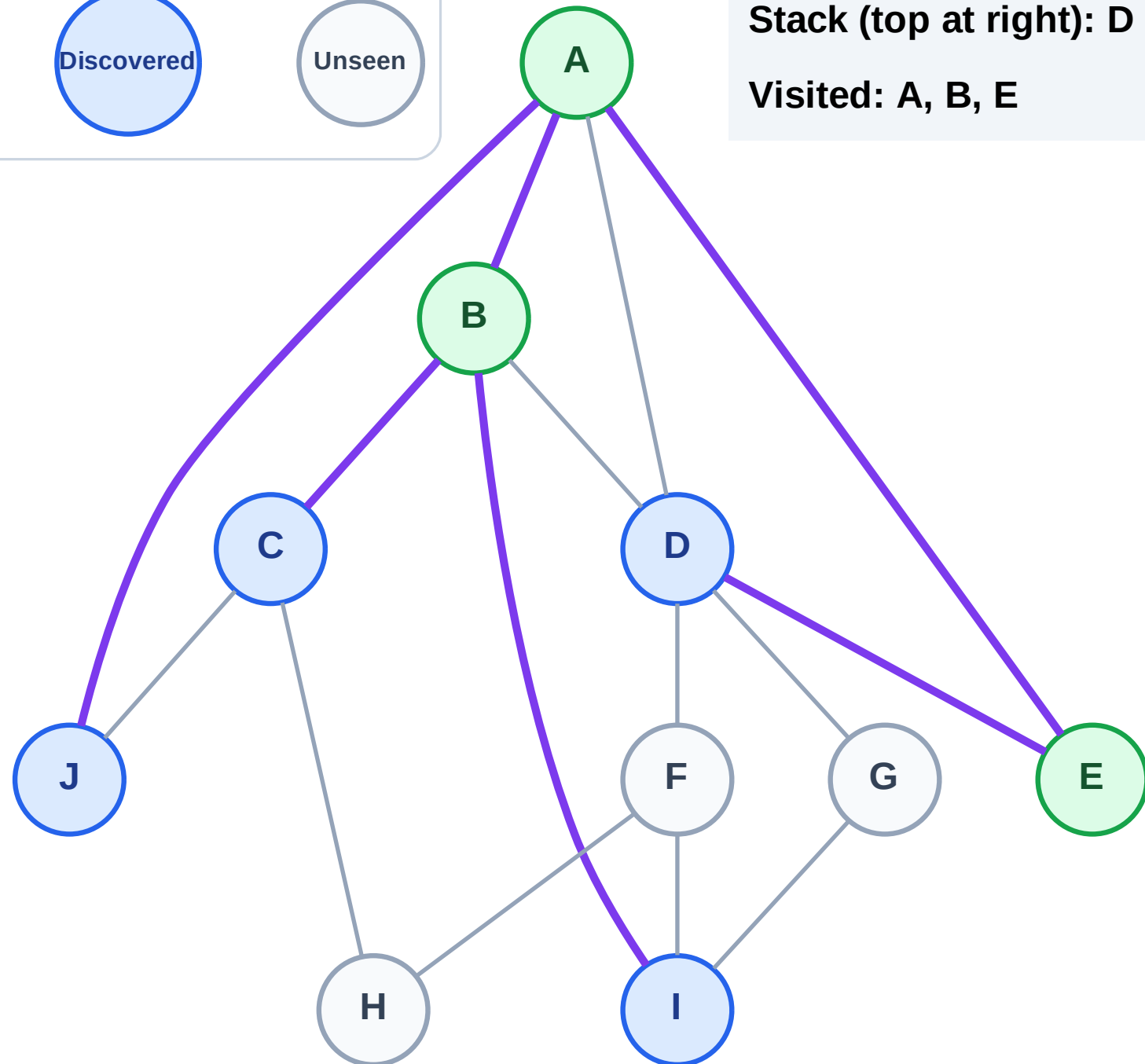
Step 7: Discover I, C from B

Legend



Stack (top at right): D | J | I | C

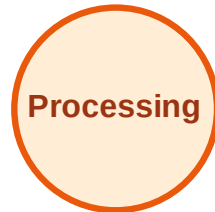
Visited: A, B, E



DFS Traversal

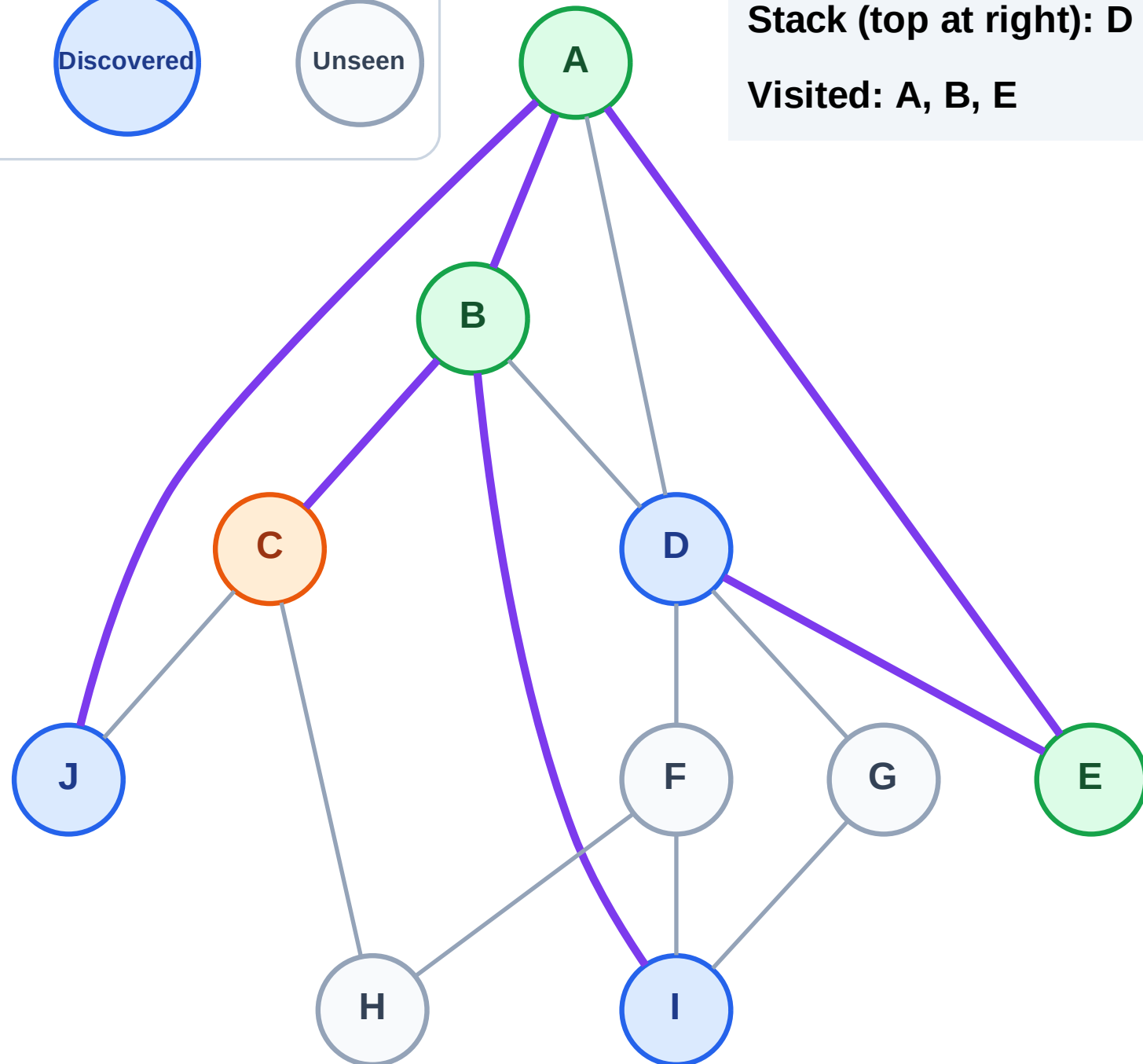
Step 8: Process node C

Legend



Stack (top at right): D | J | I

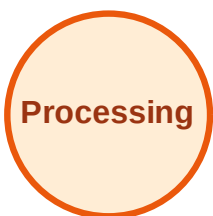
Visited: A, B, E



DFS Traversal

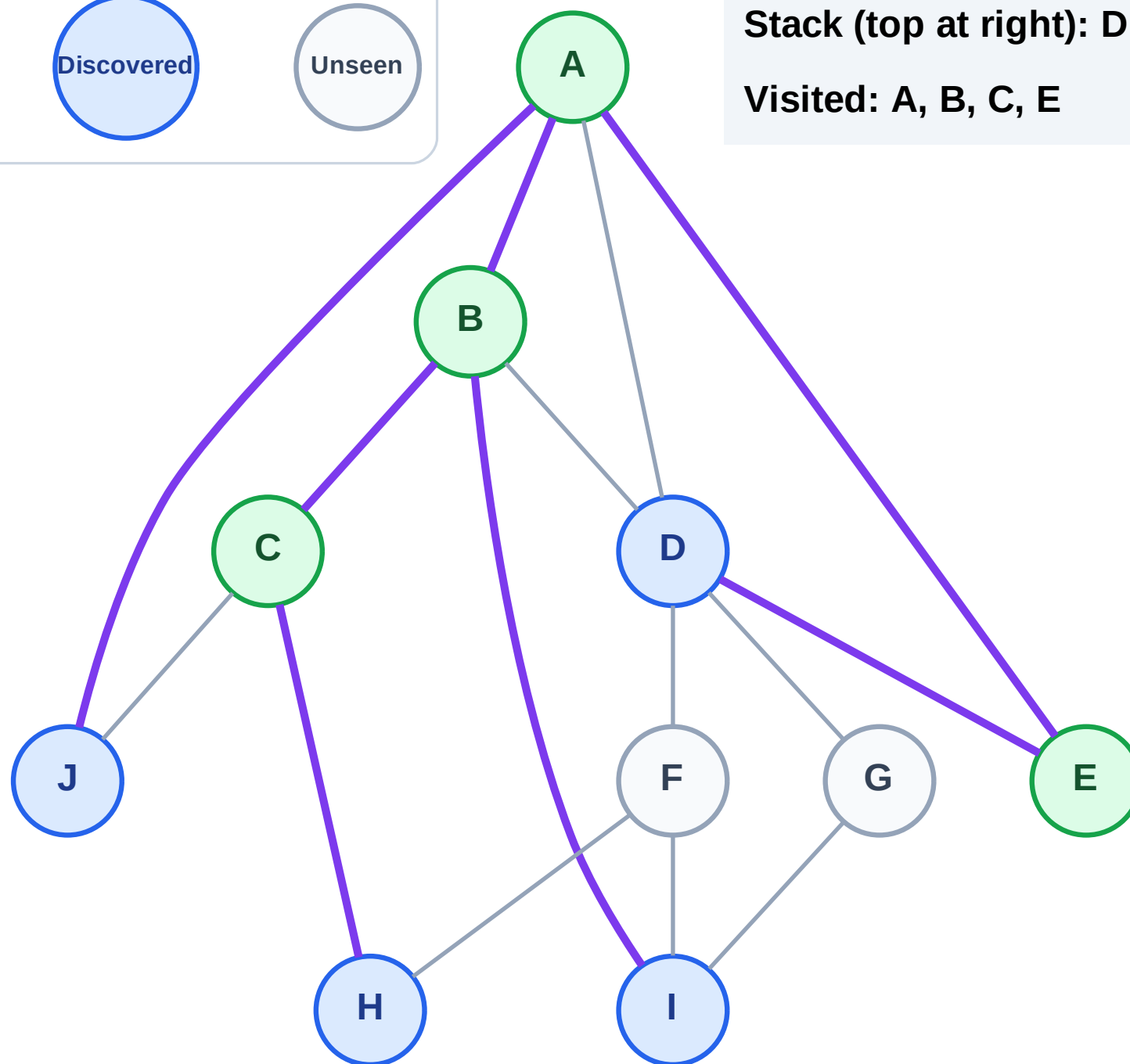
Step 9: Discover H from C

Legend



Stack (top at right): D | J | I | H

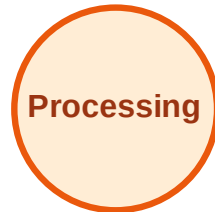
Visited: A, B, C, E



DFS Traversal

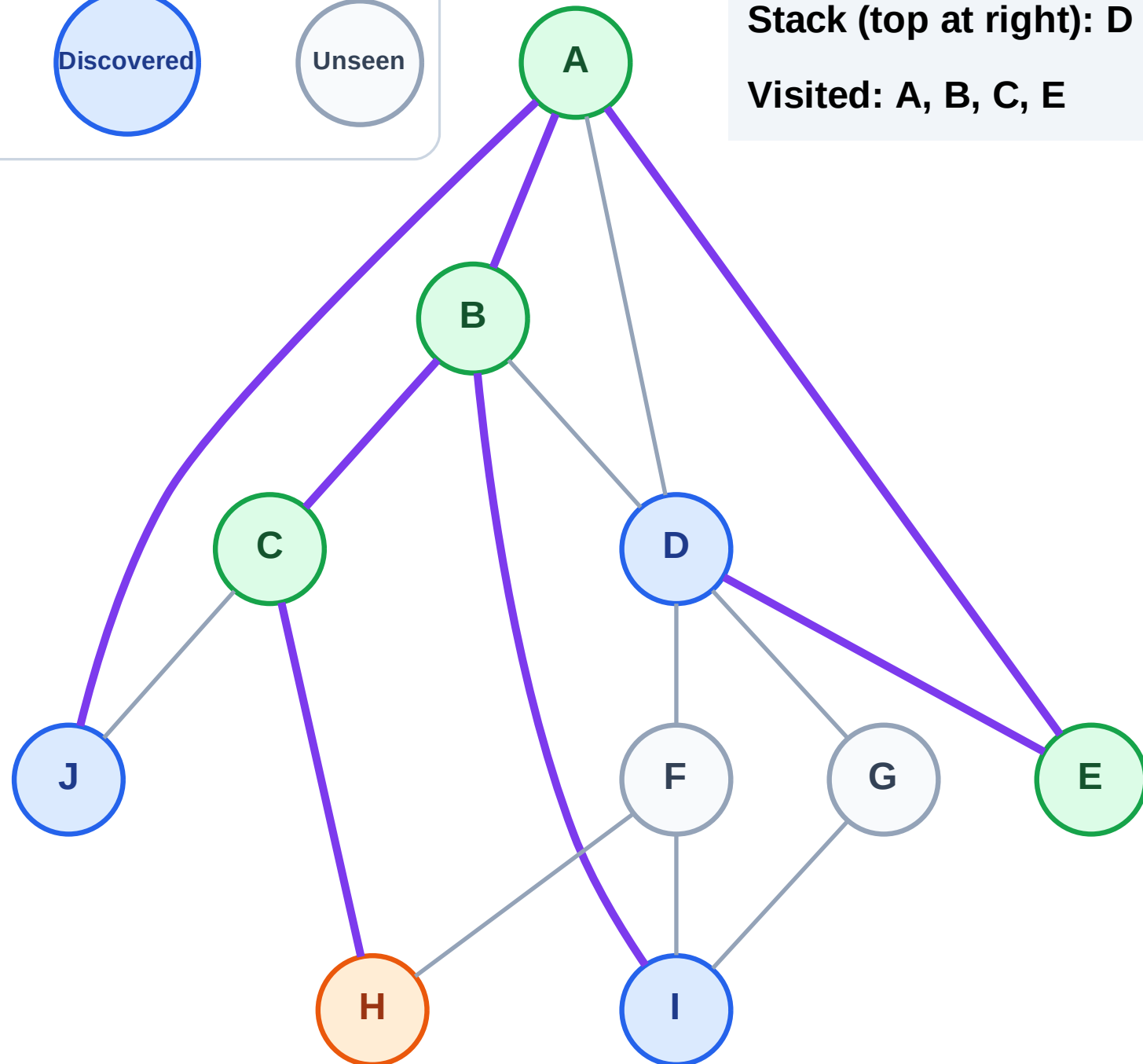
Step 10: Process node H

Legend



Stack (top at right): D | J | I

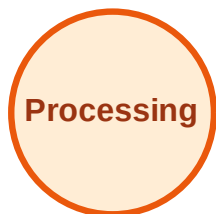
Visited: A, B, C, E



DFS Traversal

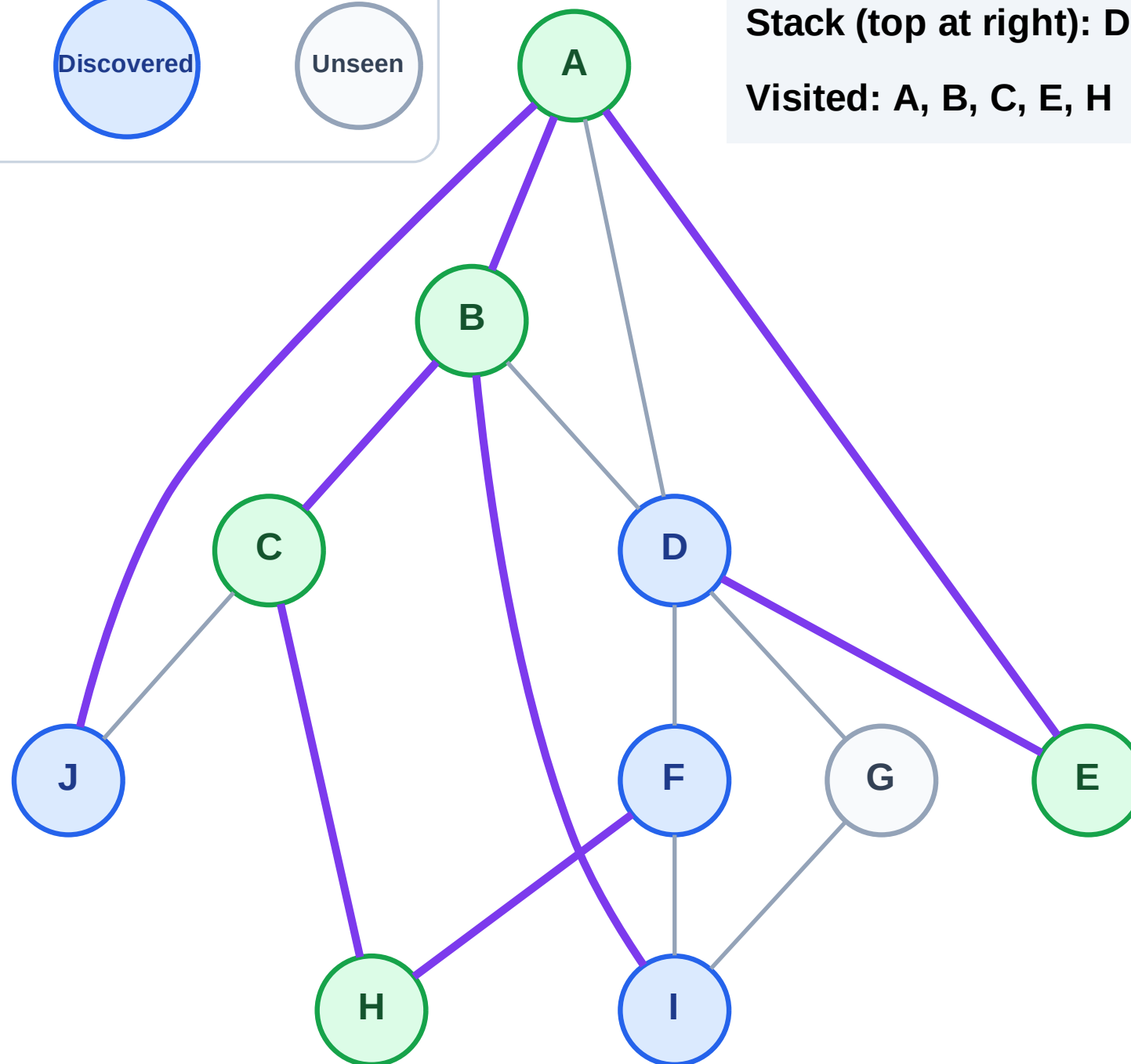
Step 11: Discover F from H

Legend



Stack (top at right): D | J | I | F

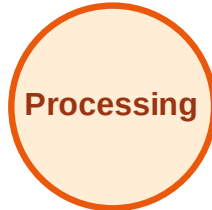
Visited: A, B, C, E, H



DFS Traversal

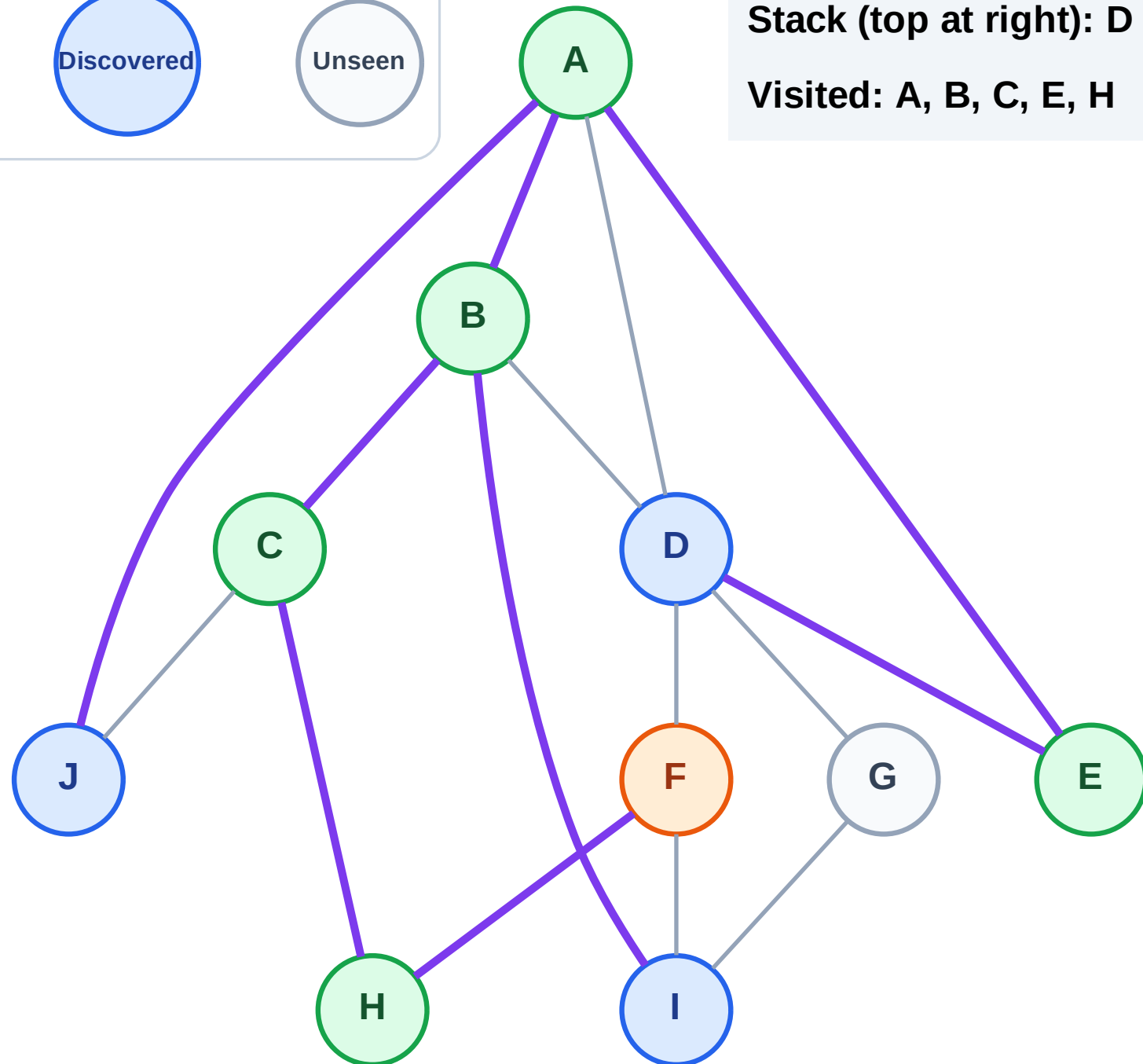
Step 12: Process node F

Legend



Stack (top at right): D | J | I

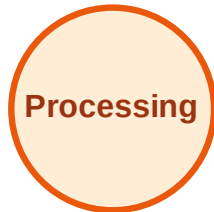
Visited: A, B, C, E, H



DFS Traversal

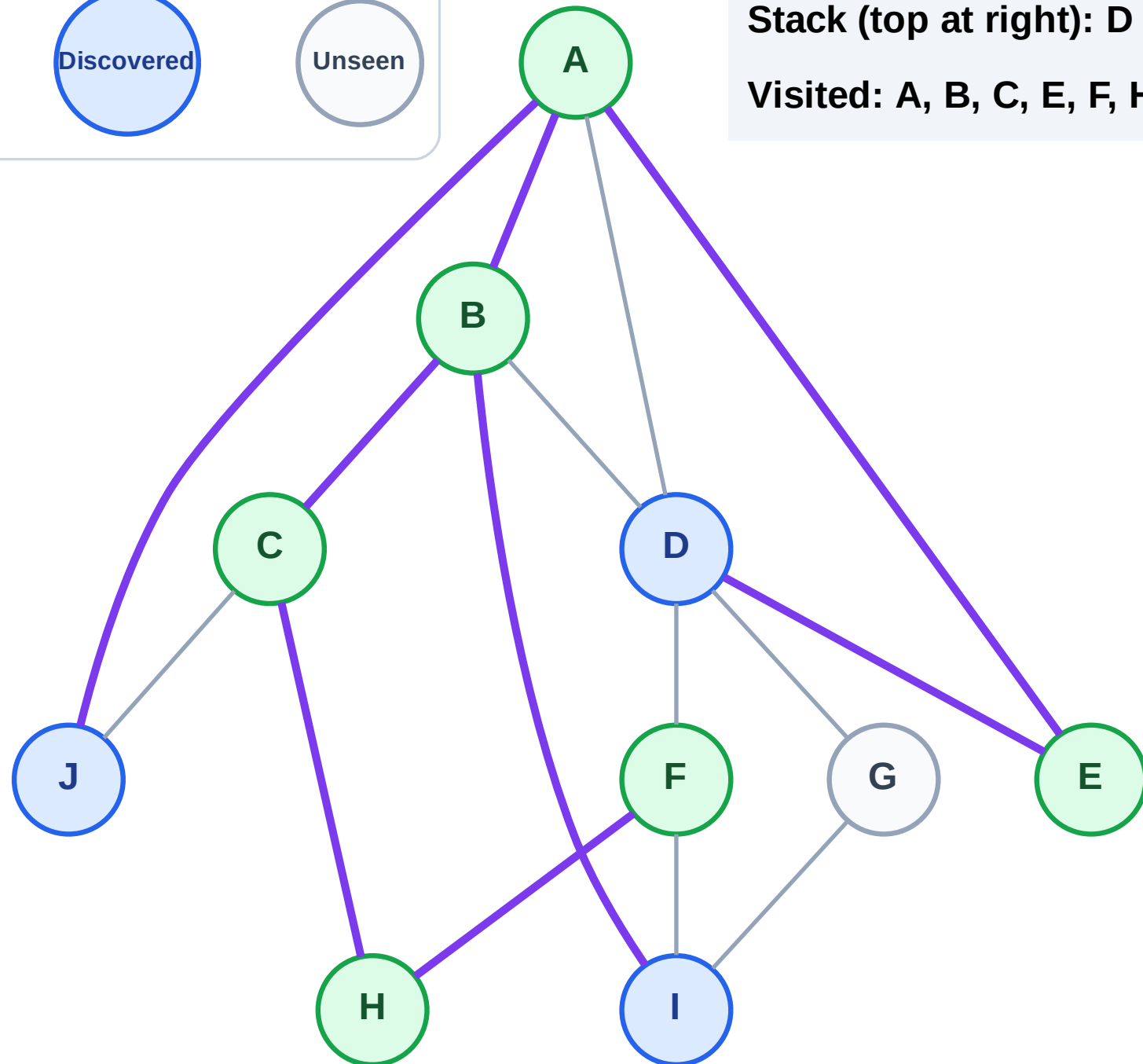
Step 13: No new neighbors from F

Legend



Stack (top at right): D | J | I

Visited: A, B, C, E, F, H



DFS Traversal

Step 14: Process node I

Legend

Complete

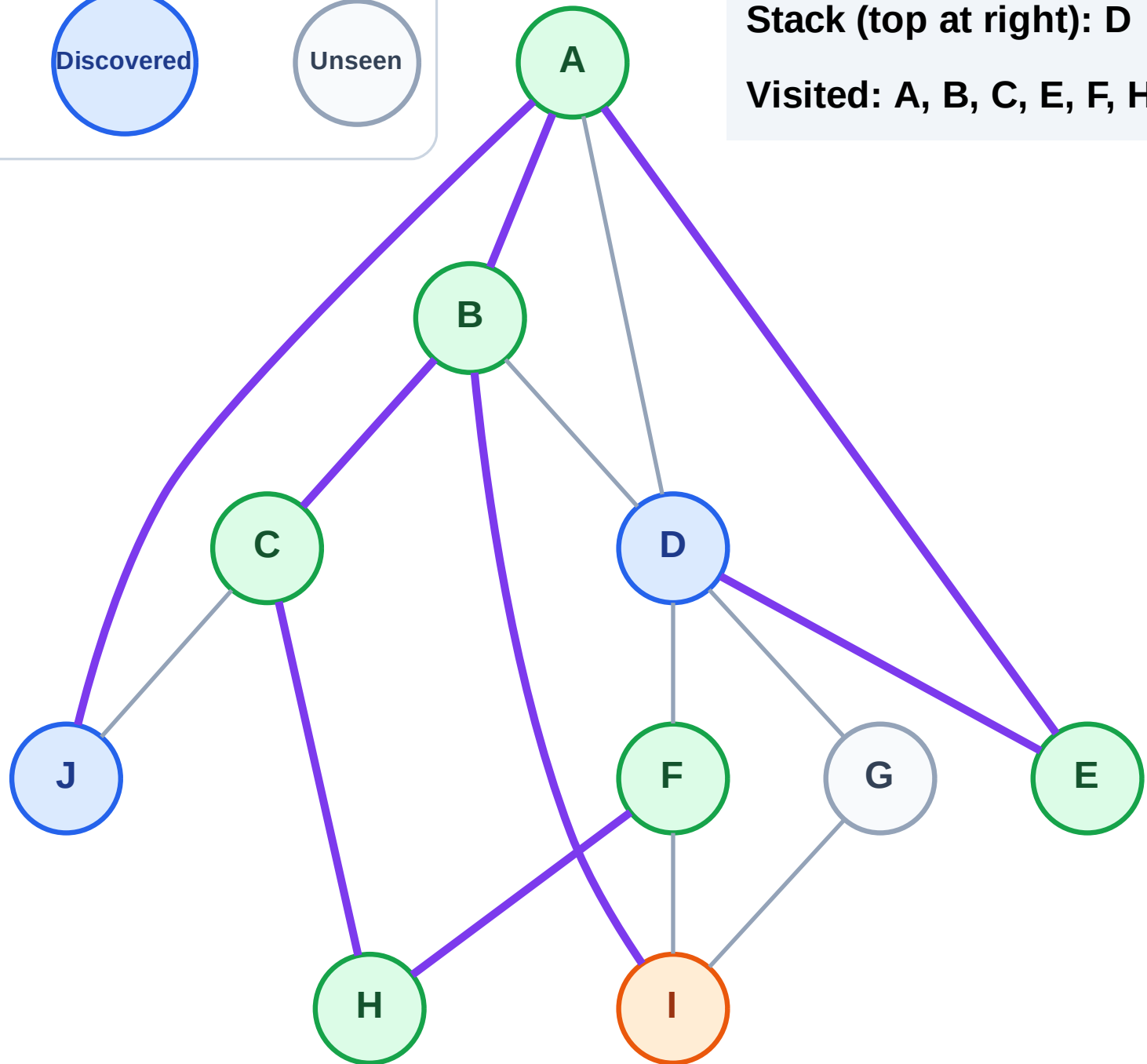
Processing

Discovered

Unseen

Stack (top at right): D | J

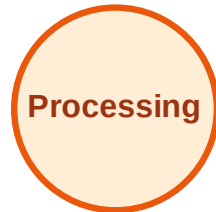
Visited: A, B, C, E, F, H



DFS Traversal

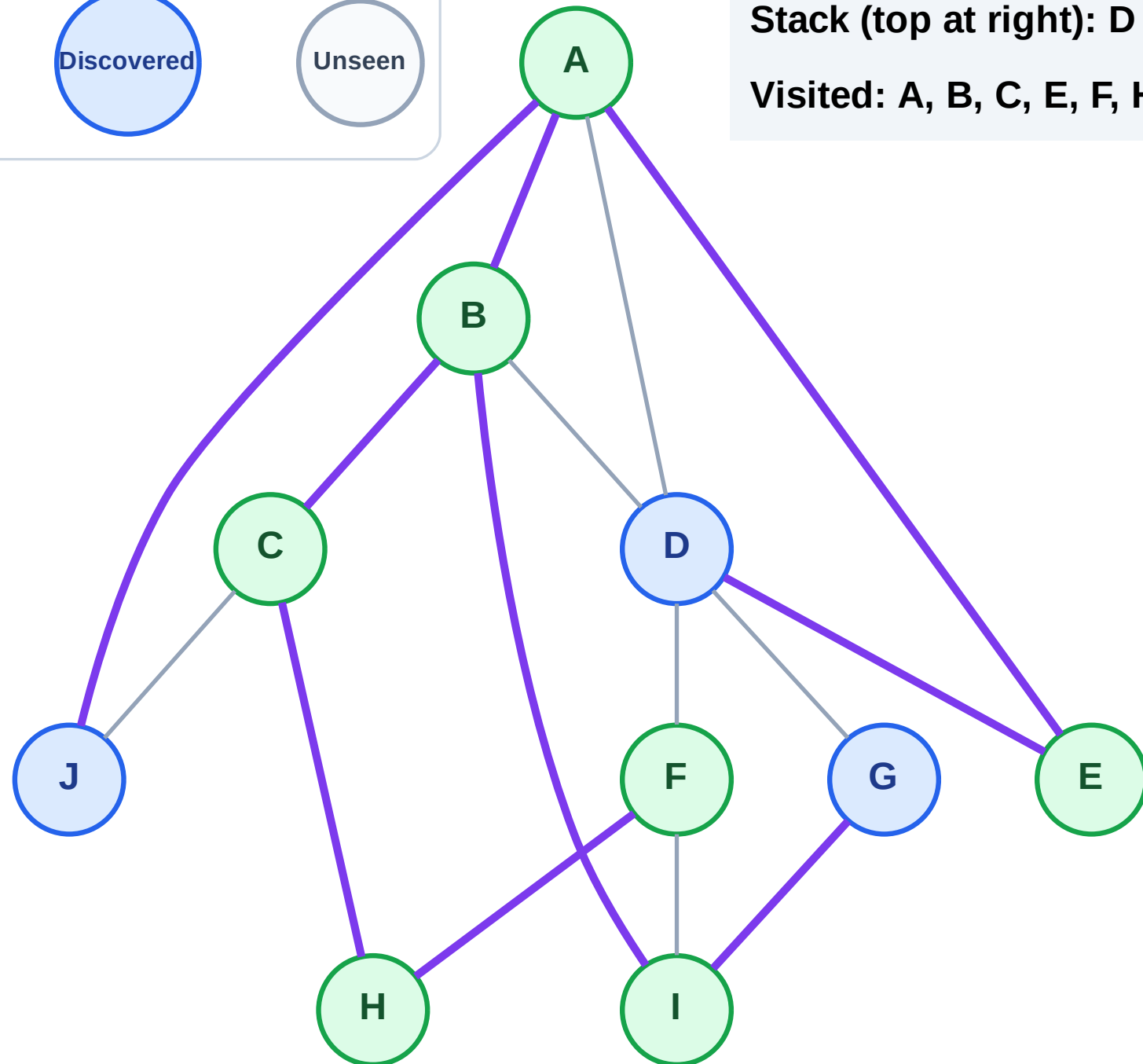
Step 15: Discover G from I

Legend



Stack (top at right): D | J | G

Visited: A, B, C, E, F, H, I



DFS Traversal

Step 16: Process node G

Legend

Complete

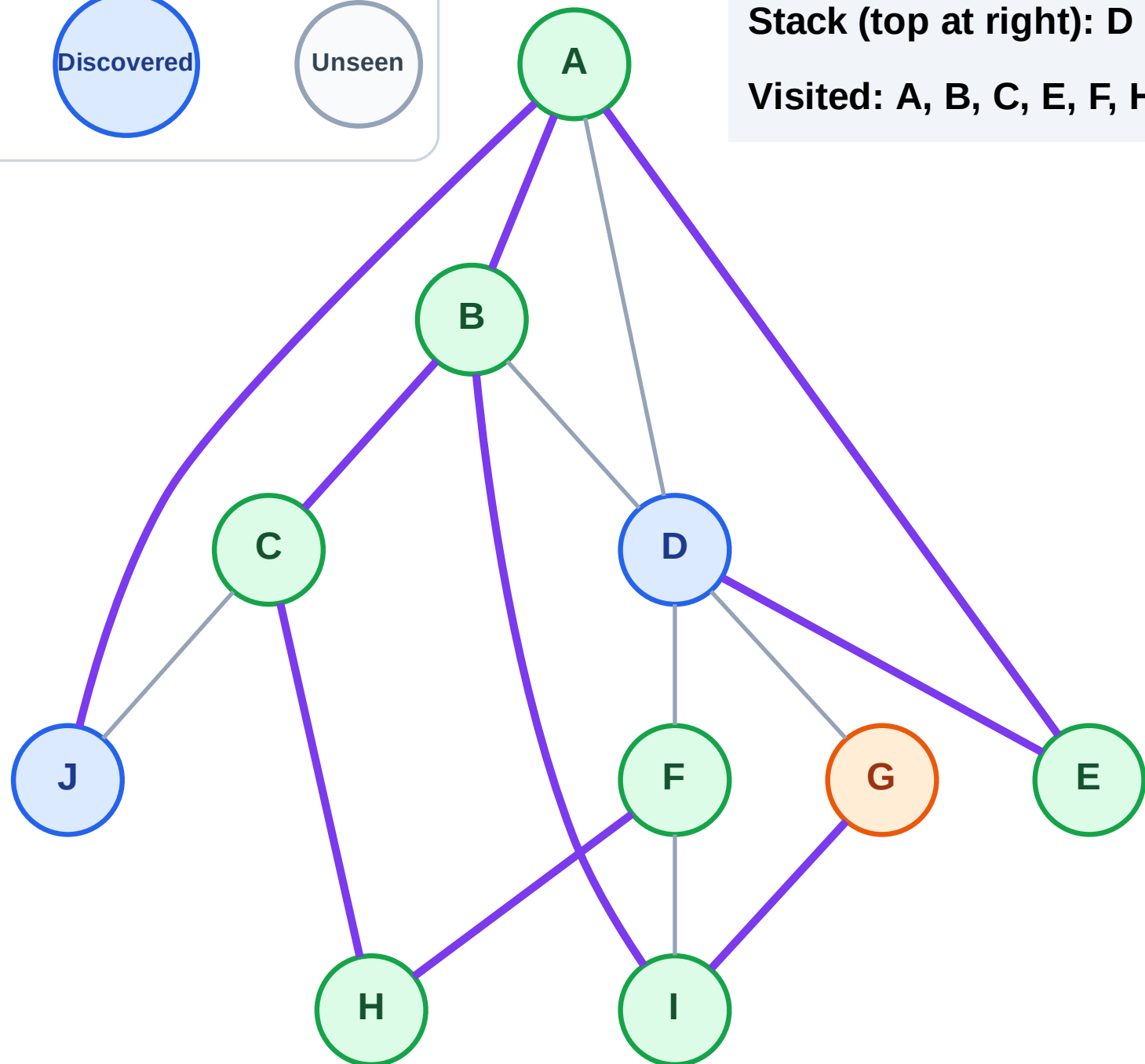
Processing

Discovered

Unseen

Stack (top at right): D | J

Visited: A, B, C, E, F, H, I



DFS Traversal

Step 17: No new neighbors from G

Legend

Complete

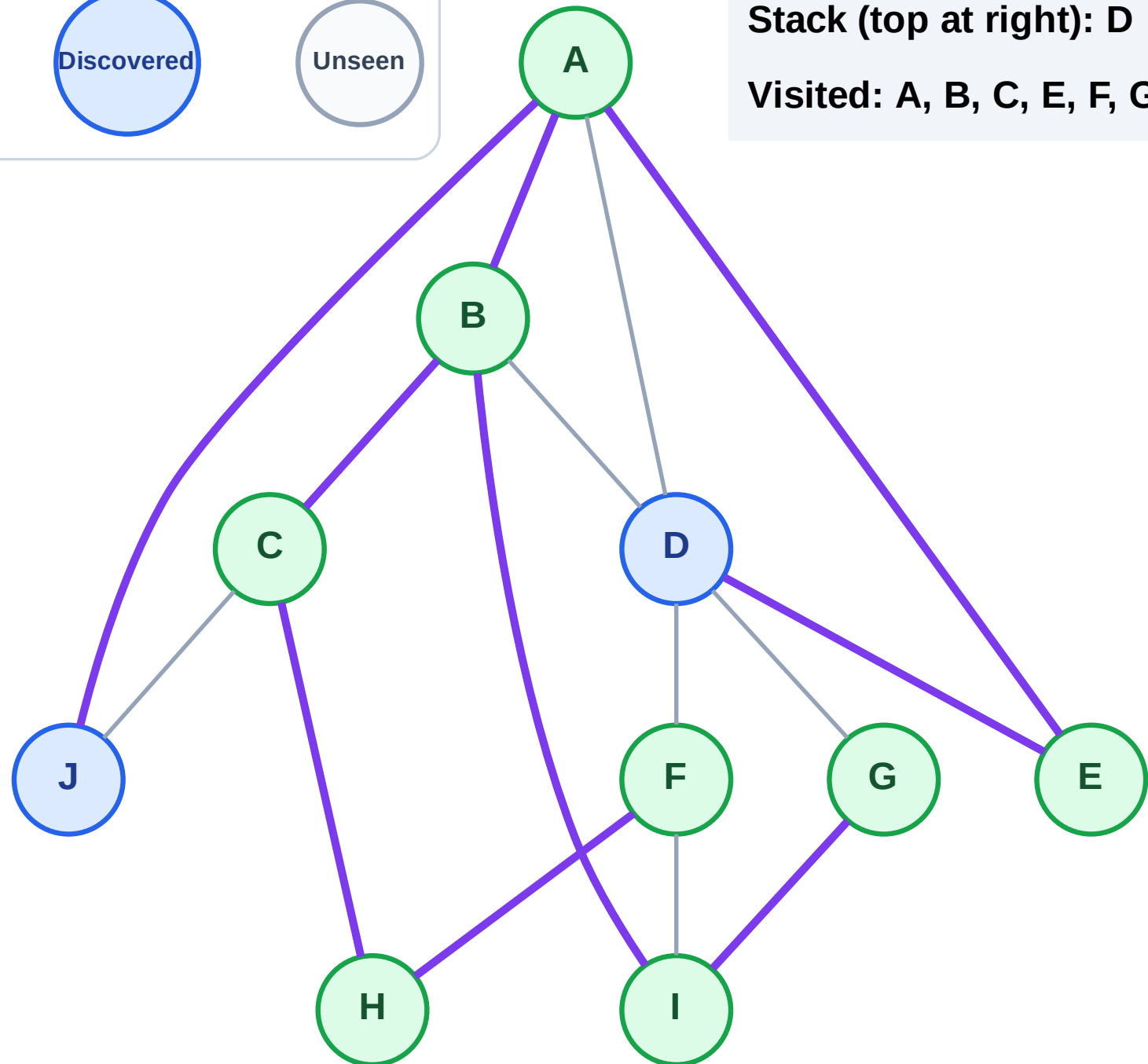
Processing

Discovered

Unseen

Stack (top at right): D | J

Visited: A, B, C, E, F, G, H, I



DFS Traversal

Step 18: Process node J

Legend

Complete

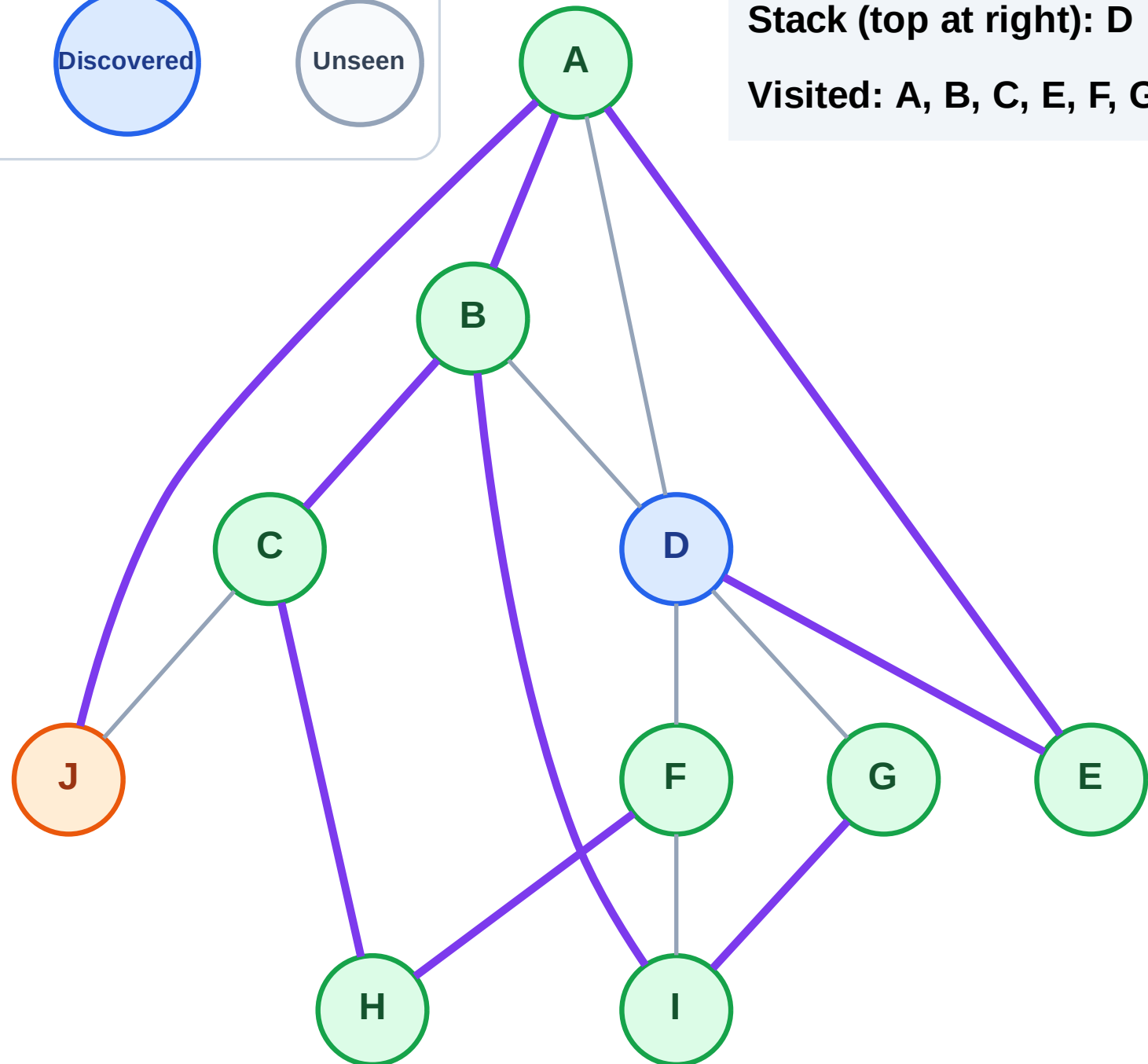
Processing

Discovered

Unseen

Stack (top at right): D

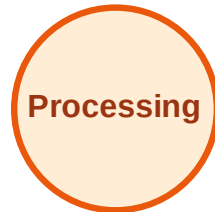
Visited: A, B, C, E, F, G, H, I



DFS Traversal

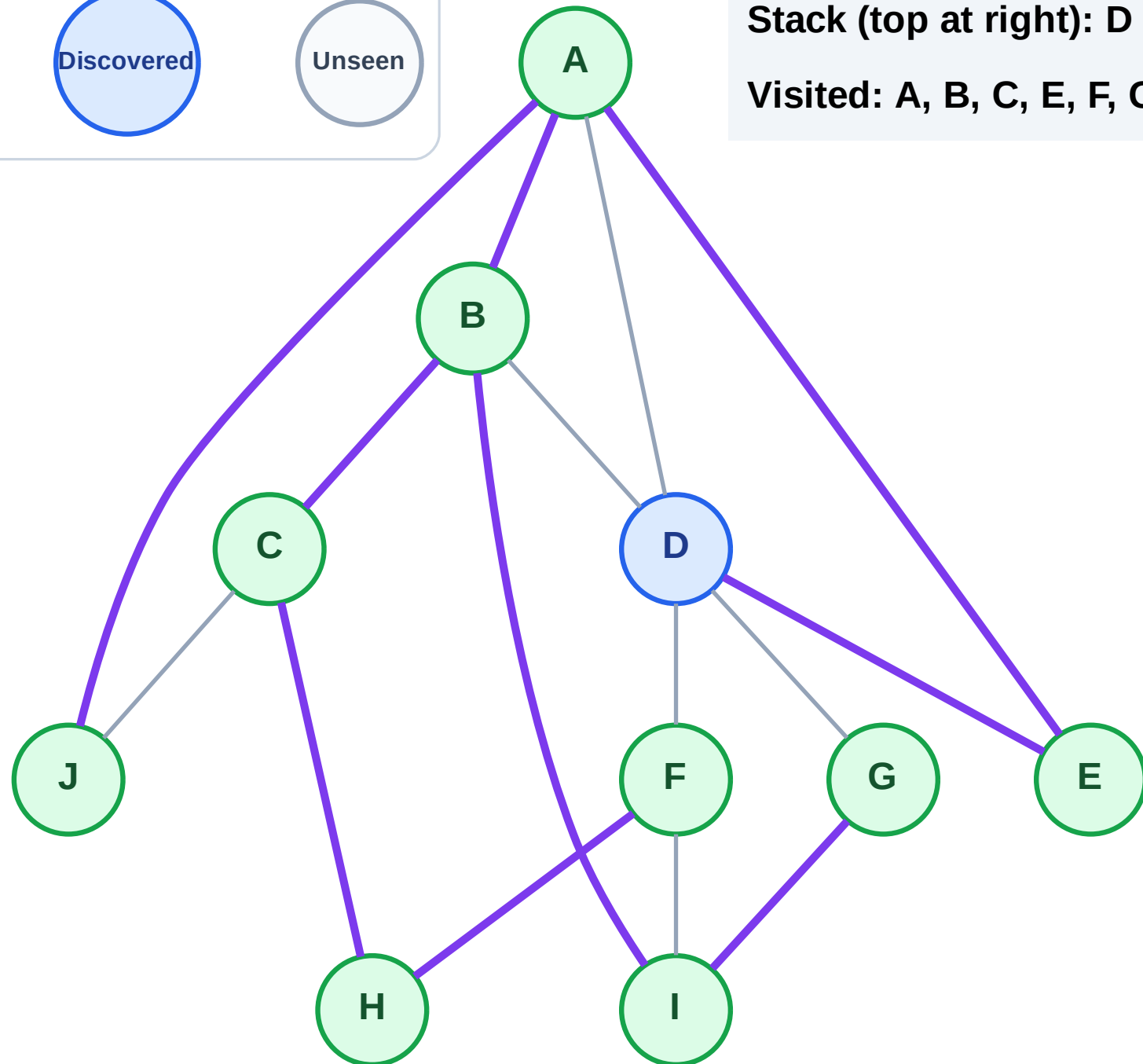
Step 19: No new neighbors from J

Legend



Stack (top at right): D

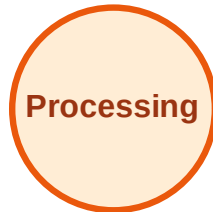
Visited: A, B, C, E, F, G, H, I, J



DFS Traversal

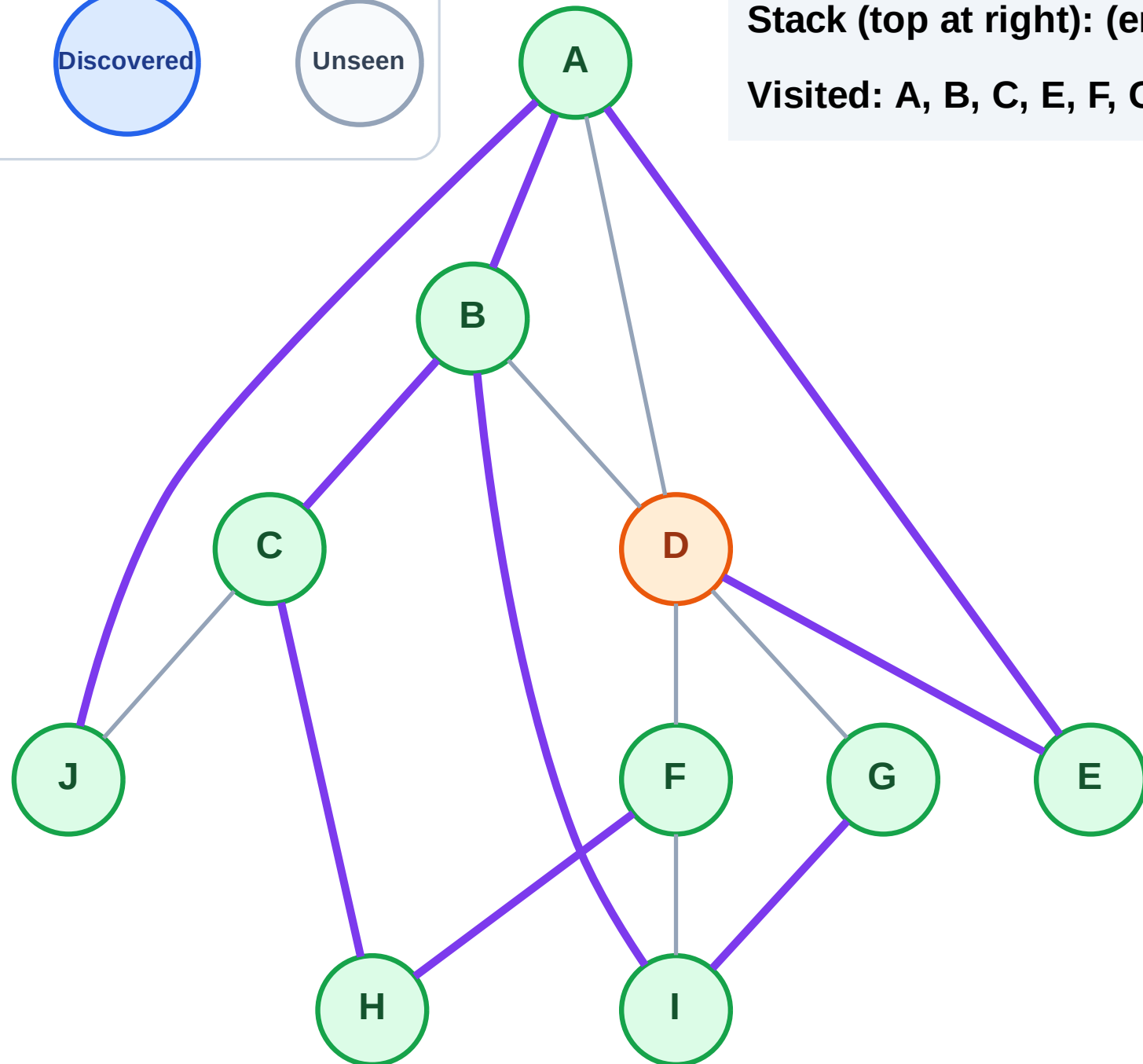
Step 20: Process node D

Legend



Stack (top at right): (empty)

Visited: A, B, C, E, F, G, H, I, J



DFS Traversal

Step 21: No new neighbors from D

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

